WIND TURBINE INSPECTION REPORT

Report Number: REU_WT_IR_002_2026
Report Date: 03-June-2026



Inspection Report: REU_WT_IR_002_2026

Company/ Organization	xxxx
Location	Wind Turbine Test Station (WTTS), Kayathar, Tamil Nadu
Wind turbine	Gamesa G97 2 MW machine
Blade-Serial #	5287, 5291, 5297
Field work dates	02-June-2026; 03-June-2026
Inspection date	xxxx

Summary

Drone-based visual inspection was conducted on three blades (5287, 5291, 5297), the nacelle, and the tower of a Gamesa G97 2 MW turbine at Kayathar, using multiple viewpoints (front/back/left/right/top/bottom) and segmented blade coverage (root → midspan → tip) across both pressure side (PS) and suction side (SS).

Overall, no confirmed structural cracks, delamination, bond-line separation, or open edges were positively identified in the captured imagery.

Most observations are surface-condition findings: rust/brown runoff staining at root-collar and hub interfaces, grease/oil staining, soiling/mottling, coating wear, and minor marks. The more significant items are leading-edge erosion at the blade tips (notably High-severity on blade 5297 PS) and rust runoff at the tower-top nacelle/yaw-ring interface, all requiring close-up verification.

Notice: Due to camera angles, stand-off distance, and lighting, certain regions (root/collar, hub interface, trailing-edge faces, crack-like indications) could not be fully resolved. Additional defects in these areas cannot be ruled out.

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1. Glossary

Terms	Definition
Wind Turbine Generator (WTG)	A generator converts wind energy into electrical power using rotor blades, hub, nacelle, drivetrain, generator, and control systems mounted on a tower, delivering electricity to the grid.
Blades	Most turbines have either two or three blades. Wind blowing over the blades causes the blades to "lift" and rotate
Blade root	A specific point on the longitudinal axis of the blade measured from the exact center of the hub. Note that the size of the hub is not the same in the various wind turbine types.
Leading edge (LE)	The leading edge of the blade, i.e. the 'nose' that is heading into the wind during operation.
Nacelle	The rotor attaches to the nacelle, which sits atop the tower and includes the gear box, low- and high-speed shafts, generator, controller, and brake. A cover protects the components inside the nacelle and large enough for a technician to stand inside while working.
Pressure Side (PS)	The PS shell is the underside during production. This side is facing the wind and is also known as the pressure side.
Suction Side (SS)	In the manufacturing the SS shell is the upper shell. It is the side of the blade which faces the tower during turbine operation.
Shell	The function of the shell is to give an aerodynamic profile.
Spar	This is the structural part of the blade. It is positioned between the shells and must resist loads and forces. The sides towards the leading and trailing edge are called the web.
Tower	Towers are made from tubular steel (shown here) or steel lattice. Because wind speed increases with height, taller towers enable turbines to capture more energy and generate more electricity.
Trailing edge (TE)	The trailing edge of the blade is the thin edge where the airflow leaves the blade during operation. Together with the main spar the trailing edge is sustaining loads to the blade.

2. Defect Classification

Severity Class	Description	Recommended Action
Critical	Extreme danger (DNV CAT 5). Structural integrity of the 47.5 m blade or drivetrain is compromised; risk of blade liberation, propagating crack or debonding under operational loads.	Stop WTG immediately and lock out. Dispatch rope-access / NDT crew for confirmation and repair before any return to service.
High	Serious damage (DNV CAT 4). Laminate exposure, lightning receptor damage, leading-edge erosion reaching structural ply, or trailing-edge bond opening that will progress if left unaddressed.	Significant defect. Schedule blade-access repair within 3 months; monitor for progression in the interim.
Medium	Damage above wear-and-tear (DNV CAT 3). Coating loss, gelcoat chipping, surface cracking or contamination with potential to develop into a structural concern on a high-tip-speed 2 MW blade.	Plan repair / recoating within 6 months and track at next inspection cycle.
Low	Cosmetic / minor non-conformance (DNV CAT 1–2). Surface soiling, staining or superficial marks with no structural significance.	No immediate action. Clean and monitor; record for trend tracking in CompliQ.

Note: This four-tier severity scale is mapped to the five-category damage convention used in DNV-recommended blade-inspection practice (CAT 1–5). For a 2 MW class machine the consequence of an unaddressed defect is materially higher than on a sub-MW turbine: the G97 blade is approximately 47.5 m long with a blade-tip speed in the order of 80 m/s, so leading-edge erosion, trailing-edge bond-line opening, lightning-receptor damage and laminate exposure escalate faster and carry greater structural and safety risk. Severity has therefore been assigned with 2 MW operational loads, tip speed and repair lead-time in mind, not only by visual appearance.

3. Introduction

3.1 Location and WTG

On 02nd and 03rd June 2026 a wind turbine inspection was conducted at Kayathar on a Gamesa G97 2MW machine.

The Gamesa G97 is a three-blade, upwind, pitch-regulated 2 MW turbine. Its principal design parameters are summarised below; these govern the structural loading, tip speed and erosion environment that frame the defect assessment in this report.

Parameter	**Value**	**Parameter**	**Value**
Rated power	2,000 kW	Rotor diameter	97 m
Blade length (approx.)	47.5 m	Swept area	7,390 m²
Tower / hub height	90 m	Rated rotor speed	15.96 rpm
Cut-in / rated / cut-out	3.0 / 11.0 / 25.0 m/s	Approx. blade-tip speed	~81 m/s
Configuration	3-blade, upwind, pitch-regulated	Blades inspected	5287, 5291, 5297

3.2 Drone Technology

The inspection was carried out with a small category hexacopter UAV, making high-resolution images of the turbine's tower, nacelle, and blades 5287, 5291, and 5297.

Company	REUDE Technology Private Limited, Chennai		
Model	REUDE-Temp850		
Category	Hexacopter small Unmanned Aerial Vehicle (UAV)		
Camera	4K 8MP Ultra HD; 6X Digital Zoom; 320-Degree Axis Rotation		
Max Endurance	25 – 30 mins	Max Range	2 km
Max Speed	15 m/s	Max Ceiling	100 m / 400 ft



Figure 1: (left) REUDE Drone with 4K Camera; (right) REUDE team at WTTS field, Kayathar

3.3 Target

The aim of this report is to inspect in detail the surface of the nacelle and blades, and documenting the status of damages wherever detected. The inspection focuses on the post-processing of UAV-captured imagery to identify visual variations. The observations are made based solely on image analysis performed using custom-developed code, limited to visible surface features, and do not constitute certified defect classification or confirmation of structural damage. Recommendations are given identified visual if any actions should be taken.

3.4 Case Data

Team

UAV Operator	Karthik, Drone Pilot
Report Reviewer	Rajesh Kumar Balaram, CTO
Report Verifier	Reginald Chellam, CEO

Unit

Location Details	Wind Turbine Test Station, Kayathar
Owner	XXX
WTG Type	Gamesa G97 2MW machine
WTG No.	XXX
Blade Serial No.	5287, 5291, 5297
Operational Hours	XXX
Production kWh	2000

Inspection Gear

Drone	REUDE MINI 850
Software	REUDE CompliQ

Weather

Weather	Sunny Sky
Wind	4 - 6 m/s
Temperature	34 - 38 °C

3.4 Inspection Details

3.4.1 Inspection Scope

The scope of this inspection includes:

- Processing a drone image of a wind turbine blade
- Segmenting the blade from the background
- Identifying visually distinct regions on the blade surface
- Highlighting these regions using bounding boxes

The scope does not include defect classification, severity grading, or root cause analysis.

3.4.2 Inspection Methodology

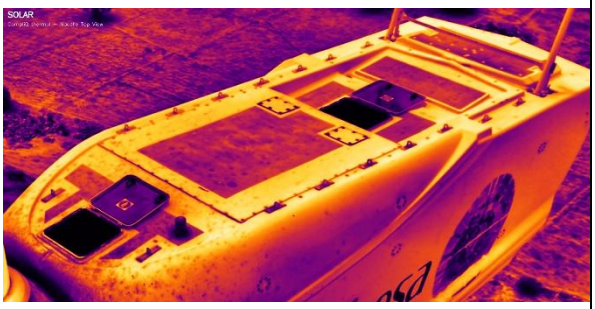
The inspection was carried out using the following steps:

1. A drone image of the wind turbine blade was used as input
2. The blade was segmented from the background using an automated segmentation model
3. The segmented blade was isolated and cropped
4. Visual intensity variations were analyzed across the blade surface
5. Regions showing noticeable visual differences were highlighted using bounding boxes

This methodology provides a preliminary indication of areas that may require further inspection.

4. Nacelle Inspection

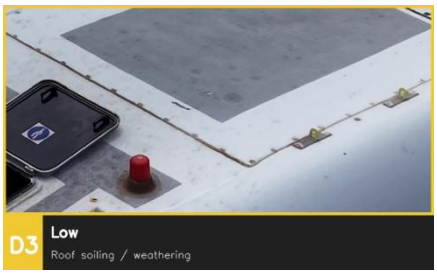
4.1 Top-View Inspection: Nacelle defects

Nacelle Top Overview	Anomaly Visualization
	

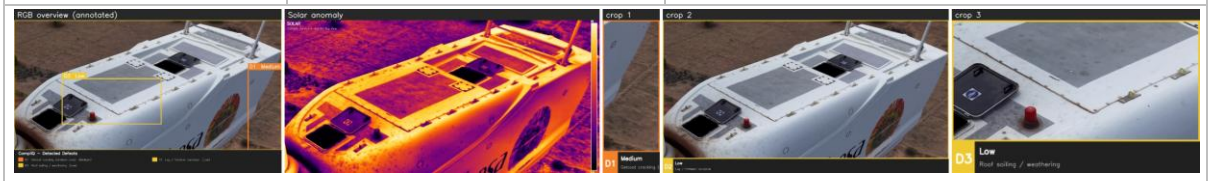
Defect Detail Evidence







Top View		3 finding(s) | max severity: Medium
Annotated View	*Anomaly Visualization*	*Defect Detail Evidence*



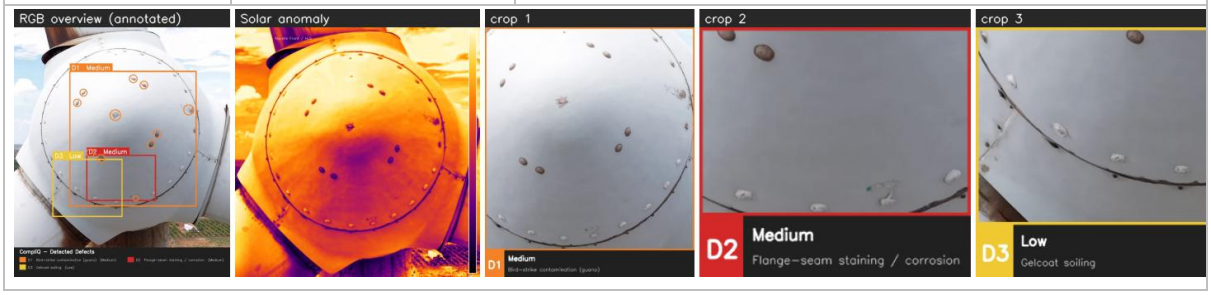
Serviceable with monitoring. The upper (roof) shell shows minor gelcoat / coating cracking near roof-mounted fixtures – to be checked close-up to rule out laminate involvement – together with general roof soiling and weathering. No fluid leaks or structural failures are evident from the top view.

Bottom View		4 finding(s) | max severity: High
Annotated View	*Anomaly Visualization*	*Defect Detail Evidence*



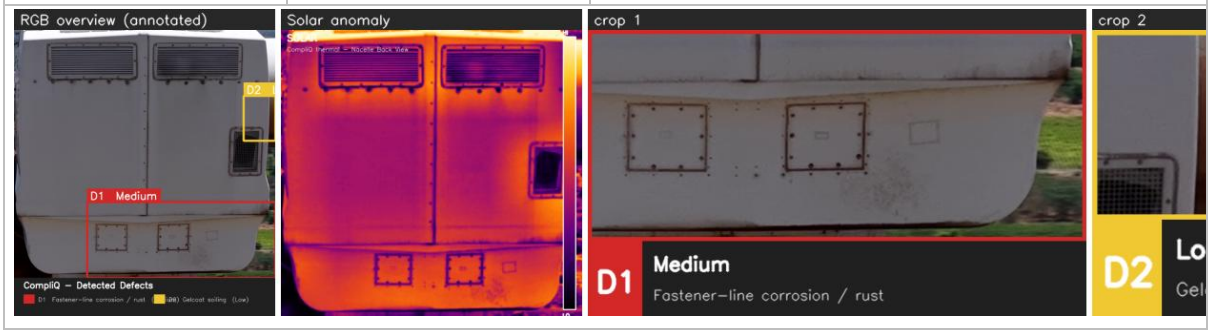
Serviceable – priority verification required. A High-severity fluid-leak / panel-seal staining indication (D1) is present and needs close-up confirmation of whether it is active; remaining findings are drip streaks (D2), biological growth (D3) and general soiling (D4), all routine-to-monitor. No structural cracks or failures confirmed from the imagery.

Front View (Nose Cone)		3 finding(s) | max severity: Medium
Annotated View	*Anomaly Visualization*	*Defect Detail Evidence*



Serviceable with monitoring. The spinner shows star / spider-web gelcoat cracking around the emblem region (check close-up to rule out laminate involvement) with surface spotting / contamination, plus flange-seam staining / corrosion at the spinner interface and light gelcoat soiling. No structural failure is evident on the nose cone.

Back View		2 finding(s) | max severity: Medium
Annotated View	*Anomaly Visualization*	*Defect Detail Evidence*



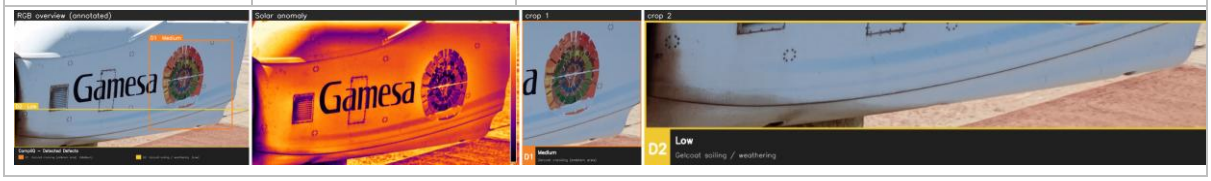
Serviceable with monitoring. The main item is rust / corrosion staining around the lower access-panel fastener lines (D1) – clean, treat and re-seal at the next window; the rest is light gelcoat soiling. The vent grilles and panel seam are design features, not defects, and the upper shell is in good condition. No structural cracks confirmed.

Left Side View		2 finding(s) | max severity: High
Annotated View	*Anomaly Visualization*	*Defect Detail Evidence*



Serviceable – priority verification required. Hub-side heavy staining / rust (D1) needs close-up confirmation of whether it is active runoff vs. weathering; lower-shell soiling (D2) is routine. The rest of the panel and the left side are relatively clean; no structural cracks confirmed.

Right Side View		2 finding(s) | max severity: Medium
Annotated View	*Anomaly Visualization*	*Defect Detail Evidence*



Serviceable with monitoring. The notable finding is star / spider-web cracking of the gelcoat over and around the emblem (D1) – likely impact or stress cracking; a close-up should confirm whether it is surface-only or reaches the laminate, with a seal / repair if confirmed. The rest is light soiling and edge staining (D2). No drivetrain leaks or structural failures are evident.

Overall Findings – Consolidated Nacelle Defects

View	ID	Observed Condition	Likely Type	Severity	Recommendation
Top View	D1	Gelcoat / coating cracking on the upper (roof) shell near roof-mounted fixtures	Gelcoat / coating stress cracking	**Medium**	Close-up inspection to rule out laminate involvement; gelcoat seal / repair if confirmed.
	D2	Roof-panel soiling across the upper shell	General atmospheric soiling / coating wear	**Low**	Monitor and clean at next cycle.
	D3	Roof soiling / weathering around roof-mounted fixtures	Atmospheric soiling / weathering	**Low**	Monitor and clean at next cycle.
Bottom View	D1	Extensive brown staining with downward streaking across the bottom shell, originating at central panel seams and fastener lines of the bolted access covers	Fluid-leak runoff (gear oil / grease / hydraulic) with panel-seal degradation	**High**	Priority check – determine if leak is active; inspect and re-seal panel covers; trace internally to gearbox / main-bearing / hydraulic source.
	D2	Vertical drip streaks and fine linear marks down the lower cylindrical component	Water / contaminant runoff; rule out hairline cracking	**Medium**	Clean and re-image at higher zoom; confirm streaks / marks are surface-only.
	D3	Greenish discolouration on the right-side shell	Biological growth (algae / lichen) from persistent damp	**Medium**	Clean and treat; identify and rectify standing-moisture / runoff source.
	D4	Widespread gelcoat soiling, weathering and fastener-area staining across the remaining surface	General atmospheric soiling / coating wear	**Low**	Monitor and clean at next cycle.

View	ID	Observed Condition	Likely Type	Severity	Recommendation
Front View	D1	Star / spider-web gelcoat cracking over and around the emblem region, with surface spotting / contamination on the spinner	Impact or stress cracking of cover gelcoat / laminate; surface contamination	**Medium**	Close-up inspection to rule out through-thickness / laminate involvement; gelcoat seal / repair if confirmed; clean spotting and check gelcoat beneath.
	D2	Flange-seam staining / corrosion at the spinner-to-nacelle interface	Fastener / sealant corrosion with rust runoff	**Medium**	Clean and treat corrosion; re-seal flange seam and check fasteners.
	D3	Gelcoat soiling, weathering and edge staining on the spinner shell	General atmospheric soiling / coating wear	**Low**	Monitor and clean at next inspection cycle.
Back View	D1	Corrosion / rust staining at lower access-panel fastener lines	Fastener corrosion / sealant breakdown with rust runoff	**Medium**	Clean and treat corrosion; re-seal access panels and check fasteners.
	D2	Gelcoat soiling / weathering on shell & panel seam	General atmospheric soiling / coating wear	**Low**	Monitor and clean at next inspection cycle.
Left Side View	D1	Localised heavy brown staining / rust runoff at the hub-side end	Weathering / grease-oil runoff; possible hub-seal trace	**High**	Priority check – inspect hub-side joints / seals; re-image to confirm if active.
	D2	Soiling / runoff staining along the lower shell and bottom edge	Runoff / atmospheric soiling band	**Medium**	Clean and re-inspect; trace runoff path.
Right Side View	D1	Star / spider-web gelcoat cracking over and around the emblem region	Impact or stress cracking of cover gelcoat / laminate	**Medium**	Close-up inspection to rule out through-thickness / laminate involvement; gelcoat seal / repair if confirmed.

View	ID	Observed Condition	Likely Type	Severity	Recommendation
	D2	Gelcoat soiling, weathering and edge staining on the shell	General atmospheric soiling / coating wear	**Low**	Monitor and clean at next inspection cycle.

Overall Findings Summary

A total of 16 external-surface indications were identified across the six nacelle views (0 Critical, 2 High, 8 Medium, 6 Low). The two High-severity items are fluid-leak / runoff staining at the bottom-shell panel seams and fastener lines (Bottom D1) and heavy staining / rust at the hub-side end (Left D1); both are the priority items and require close-up confirmation of whether the leak / runoff is active, tracing to the gearbox / main-bearing / hydraulic / hub-seal source, and re-sealing of panel covers.

The Medium-severity findings cluster into three themes: (1) drip streaks / fine linear marks to be cleaned and re-imaged to rule out hairline cracking; (2) corrosion / rust staining at fastener lines and flange seams (back and front) requiring cleaning, treatment and re-sealing; and (3) star / spider-web gelcoat cracking at the emblem regions (front, right) and roof gelcoat cracking (top), all to be checked close-up to rule out laminate involvement – plus biological growth (algae) and a runoff soiling band. Low-severity items are general gelcoat soiling and weathering. No structural cracks, panel breaches or through-thickness damage were confirmed from the imagery.

Conclusion

- The external nacelle condition is serviceable based on the drone imagery; no confirmed structural failures, panel breaches or through-thickness damage were observed.
- The dominant maintenance theme is fluid-leak / runoff and corrosion staining at panel seams, fastener lines and the hub-side end (2 High, several Medium), which should be confirmed close-up and traced to source.
- Gelcoat star / spider-web cracking at the emblem and roof regions (Medium) should be checked close-up to rule out laminate involvement.
- Continued operation is acceptable provided the priority verification (active leak / runoff confirmation, panel re-seal, corrosion treatment) and routine cleaning are completed within the next planned maintenance window.

Recommendations

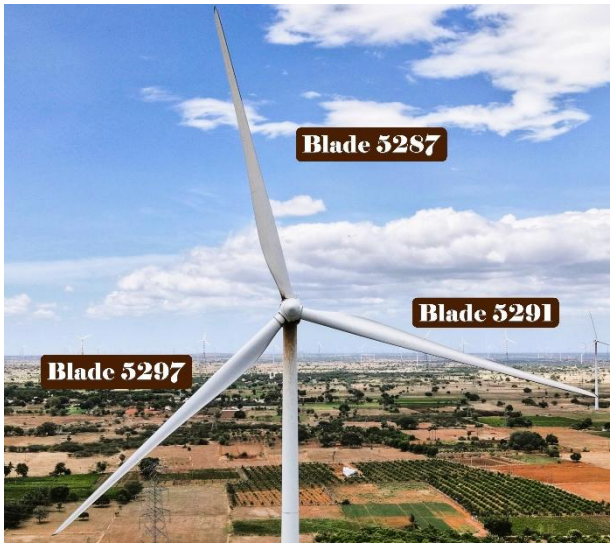
1. Priority re-fly / close-up of the two High items (Bottom D1 fluid-leak staining; Left D1 hub-side staining / rust) to confirm if active; trace internally to gearbox / main-bearing / hydraulic / hub-seal and re-seal panel covers.
2. Clean and treat corrosion / rust staining at the back and front fastener lines and flange seams; re-seal access panels and check / replace fasteners.
3. Close-up gelcoat-crack assessment at the front and right emblem regions and the top roof panel; gelcoat seal / repair if laminate involvement is confirmed.
4. Clean biological growth (algae) and rectify the standing-moisture / runoff source; clean runoff soiling bands.
5. Apply a clean-and-reinspect approach to separate soiling from true coating loss / corrosion; maintain a standardized multi-view drone checklist for consistent trend tracking in CompliQ.

5. Blade Inspection

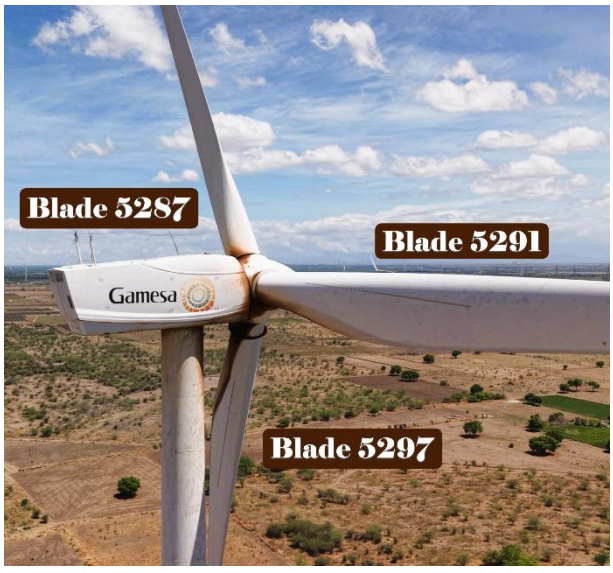
This chapter presents the drone-based visual inspection results for three wind turbine blades: **Blade 5287, Blade 5291,** and **Blade 5297**. High-resolution images were captured to assess both sides of wind turbine blades:

Side	What it is	High-risk zones	Common Defects observed
Suction Side	Leeward surface; lower pressure	Trailing edge (TE), bond lines, tip region, near root	TE cracks, bond line separation, surface bulging, rain erosion, delamination, damage marks
Pressure Side	Wind-facing surface; higher pressure	Leading edge (LE), mid-span, near root, drain paths	LE erosion, particle impact pitting, paint/coating peel, surface cracks, discoloration

Suction Side

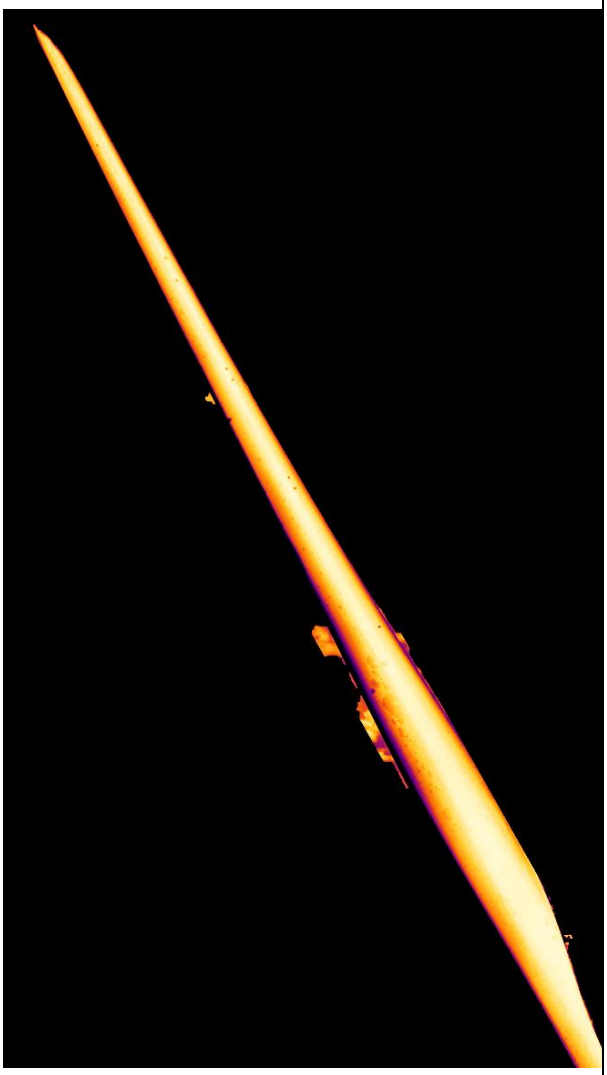


Pressure Side

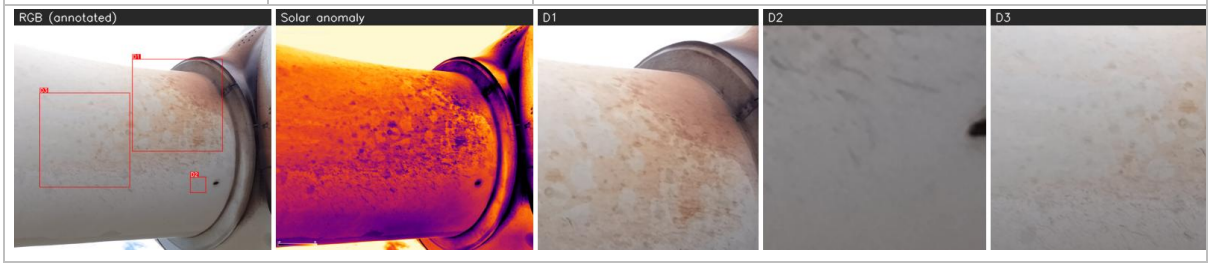


5.1 Blade 5291

5.1.1 Pressure Side - Segment-by-Segment Findings

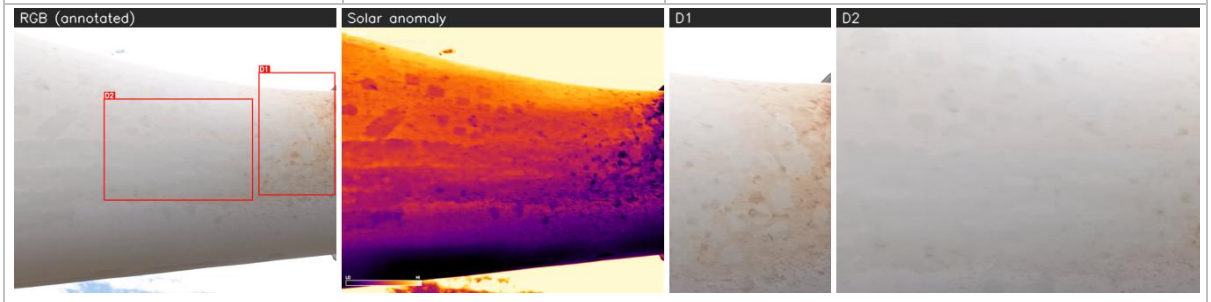
Pressure Side (PS)	
	

PS Segment — 0% (Blade Root Hub)		3 finding(s) | max severity: Low
Annotated View	*Anomaly Visualization*	*Defect Detail Evidence*



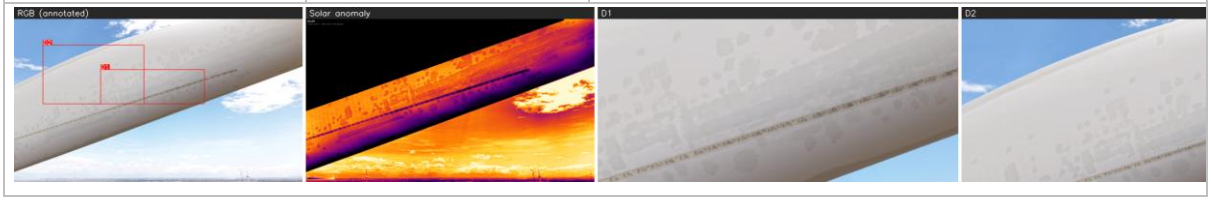
Blade root barrel and the root-collar / pitch-bearing interface were inspected. The pressure-side surface is broadly intact. The dominant finding is tan/brown discoloration and diffuse soiling consistent with grease/oil runoff migrating from the hub / pitch-bearing region, together with one small localized dark spot. No cracks, gel-coat fracture, or edge separation are evident. Root-collar hardware (bolt band, bearing seal) appears intact in the captured view.

PS Segment — 20% (Root to Mid-Span)		2 finding(s) | max severity: Low
Annotated View	*Anomaly Visualization*	*Defect Detail Evidence*



Generally uniform pressure-side surface. Tan/brown staining is present toward the root-side edge with diffuse circular mottling / light soiling across the panel. Indications are consistent with contamination rather than structural damage. No cracks or laminate exposure observed.

PS Segment — 40% (Mid-Span)		2 finding(s) | max severity: Low
Annotated View	*Anomaly Visualization*	*Defect Detail Evidence*



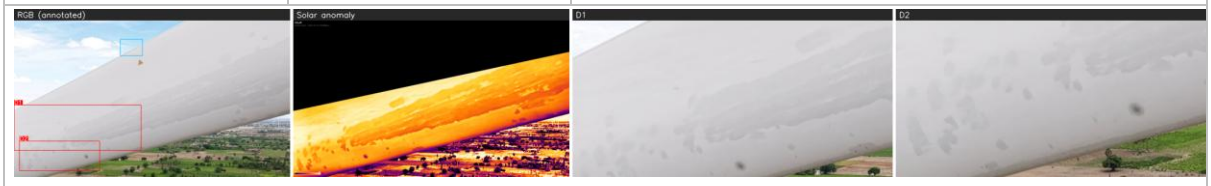
Midspan PS panel with the trailing-edge bond / serration line visible and appearing continuous and intact. Light patchy discoloration / mottling is present (gel-coat unevenness or cleaning marks). No open crack or debond is confirmed; the trailing-edge line is treated as a routine verification item.

PS Segment — 60% (Mid-Span)		2 finding(s) | max severity: Low
Annotated View	*Anomaly Visualization*	*Defect Detail Evidence*



Comparable to the 40% segment. The trailing-edge bond / serration line is continuous, with broad patchy grey discoloration across the PS panel. No crack-like propagation pattern is evident.

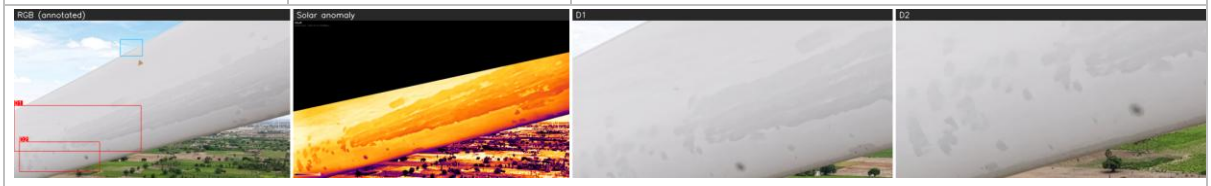
PS Segment — 70% (Mid-Span)		2 finding(s) | max severity: Low
Annotated View	*Anomaly Visualization*	*Defect Detail Evidence*



Markers: M1: triangular zone / position marker — existing label, not a defect.

Pressure-side surface shows grey streaking / discoloration banding along the trailing-edge side and small dark soiling spots near the lower edge. A triangular position / zone marker (existing label, not a defect) is present and is boxed in cyan for reference. No cracks observed.

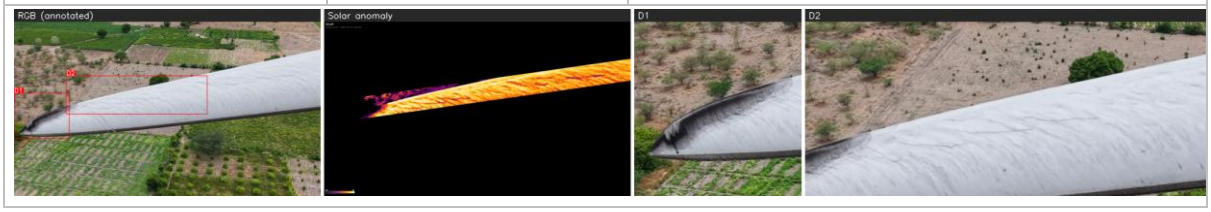
PS Segment — 80% (Mid-Span to Tip)		2 finding(s) | max severity: Low
Annotated View	*Anomaly Visualization*	*Defect Detail Evidence*



Markers: M1–M3: triangular zone / position markers — existing labels, not defects.

Outboard PS panel with patchy surface discoloration / mottling and trailing-edge darkening / streaking. Three triangular position / zone markers (existing labels, boxed in cyan) are present. No open edge or crack is confirmed; the trailing edge appears continuous.

PS Segment — 100% (Blade Tip)		2 finding(s) | max severity: Medium
Annotated View	*Anomaly Visualization*	*Defect Detail Evidence*



Tip region. This segment carries the most significant finding on the blade: a darkened, vertically streaked band at the leading-edge tip consistent with leading-edge erosion / coating wear, together with longitudinal surface streaking along the outboard leading edge. The tip / edge is a high-stress, high-risk zone. No open crack is confirmed in the captured view, but close-up verification is warranted. The trailing edge appears continuous.

Overall Findings – Consolidated PS Defects

Span	ID	Observed Condition	Category	Severity	Recommendation
0%	D1	Broad tan/brown discoloration band on the PS near the root collar	Grease/oil runoff – surface contamination / coating stain	**Low**	Clean and re-image at higher zoom to confirm no coating loss beneath the stain; trace grease source at the hub / pitch-bearing seal.
	D2	Small isolated dark spot on the white surface near the collar	Drain-hole / impact mark or localized soiling	**Low**	Confirm whether drain hole or surface mark; clean and monitor at next cycle.
	D3	Diffuse mottled soiling across the root barrel	Dirt / soiling	**Low**	Routine cleaning; no further action required.
20%	D1	Tan/brown staining patch toward the root-side edge	Grease/oil or coating stain	**Low**	Clean and re-image; check upstream grease migration from the root/hub.
	D2	Diffuse circular mottling / light soiling across the panel	Dirt / soiling / gel-coat mottling	**Low**	Clean; monitor at next inspection.
40%	D1	Trailing-edge bond / serration line (long linear feature)	TE bond-line / serrated edge – verification item	**Low**	Verify continuity at higher zoom; confirm no bond-line opening or serration loss.

Span	ID	Observed Condition	Category	Severity	Recommendation
	D2	Patchy surface mottling / discoloration on the panel	Gel-coat unevenness / soiling	**Low**	Clean and re-image.
60%	D1	Trailing-edge bond / serration line	TE bond-line – verification item	**Low**	Verify continuity at higher zoom.
	D2	Patchy grey discoloration / mottling	Surface contamination / gel-coat unevenness	**Low**	Clean and re-image.
70%	D1	Grey streaking / discoloration band along the trailing-edge side	Runoff stain / surface contamination / early coating wear	**Low**	Clean and re-image with raking light to rule out coating wear.
	D2	Small dark spots / localized soiling near the lower edge	Dirt / soiling / minor marks	**Low**	Clean; monitor.
80%	D1	Patchy surface discoloration / mottling	Surface contamination / gel-coat unevenness	**Low**	Clean and re-image.
	D2	Trailing-edge darkening / streak	TE soiling / early coating wear	**Low**	Verify TE continuity at higher zoom; clean.
100%	D1	Darkened, vertically streaked band at the leading-edge tip	Leading-edge erosion / tip coating wear	**Medium**	Close-up inspection to confirm erosion depth; schedule LE repair / recoat and apply LE protection (tape/coating) if gel-coat or laminate exposure is confirmed.
	D2	Longitudinal surface streaking / erosion banding along the outboard LE	Surface erosion-related wear / streaking	**Low**	Clean and re-image; monitor erosion progression at the next cycle.

Overall Findings Summary

Across the seven pressure-side stations, 15 indications were recorded — 14 Low and 1 Medium, with no High or Critical findings. The pressure side is dominated by surface-condition features: grease/oil runoff staining and diffuse soiling at the root and inboard span, patchy gel-coat mottling through the midspan, and grey streaking on the outboard panels. The trailing-edge bond line at 40–60% appears continuous. The only Medium item is leading-edge erosion at the tip, which warrants close-up confirmation of gel-coat or laminate exposure. No structural cracks, delamination, open edges or bond-line separation were observed. The blade is serviceable and may continue operating, subject to root-stain tracing and tip verification at the next window.

Conclusion

The PS of blade 5291 is serviceable: no confirmed cracks, delamination, open edges or bond-line separation in the captured imagery. Most findings are surface-condition (root grease/oil staining, soiling, gel-coat mottling, outboard streaking). The most significant item is Medium-severity leading-edge erosion at the tip (100% D1).

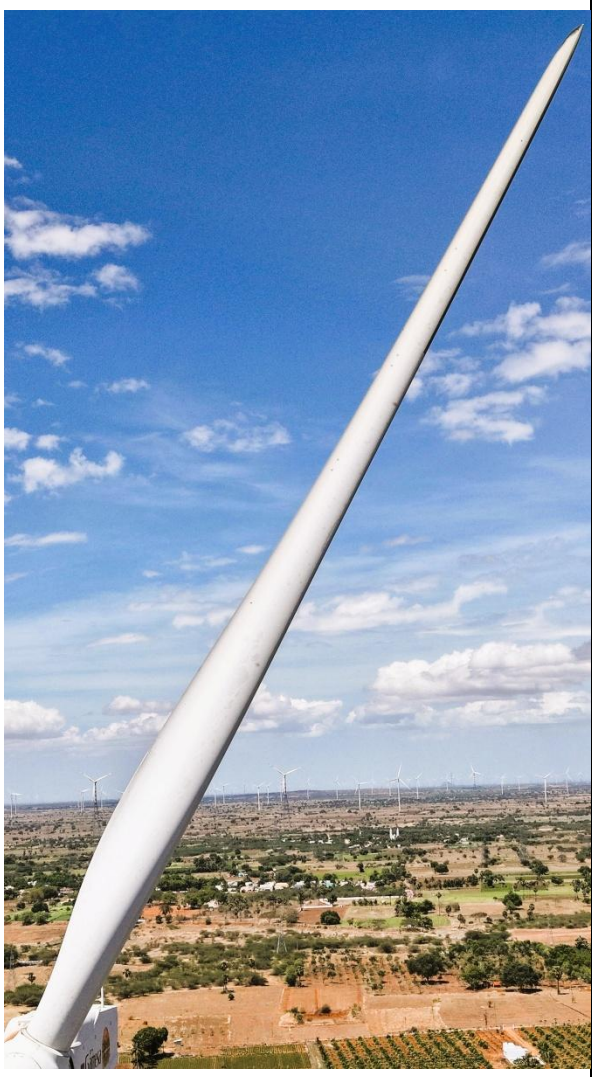
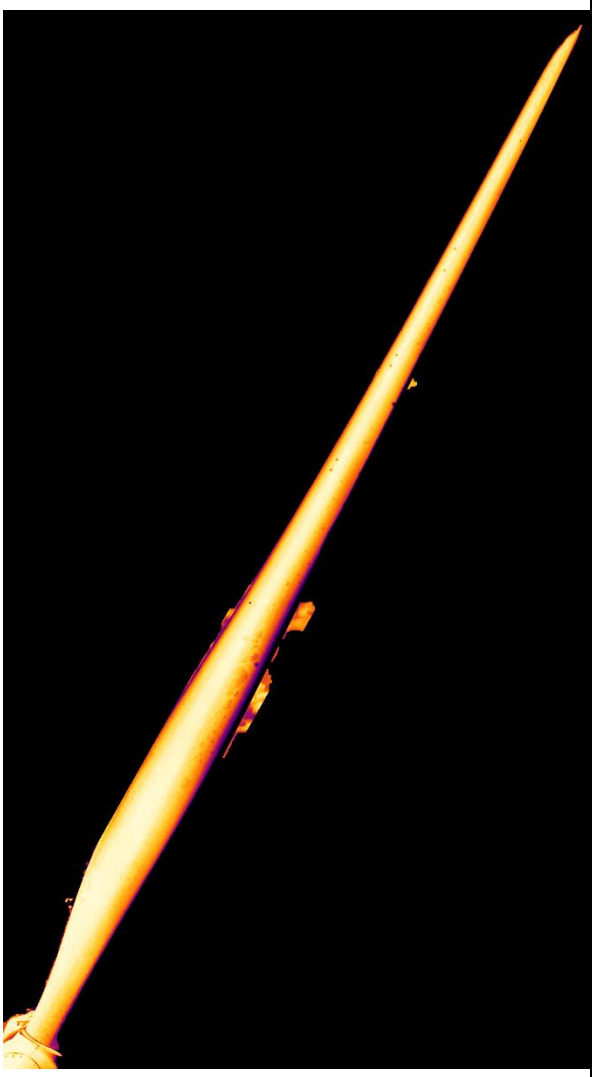
Recommendations

Continued operation is acceptable subject to the actions below.

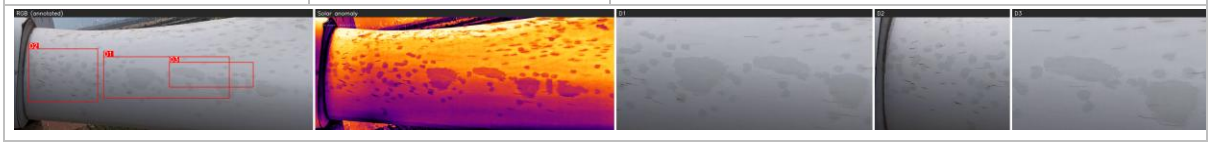
1. Priority verification re-flight (higher zoom, 2 angles) of tip LE erosion (100% D1); schedule LE repair/recoat + LE protection if gel-coat/laminate exposure is confirmed.
2. Clean-and-reinspect root grease/oil staining (0–20%); trace migration at hub/pitch-bearing seal.
3. Verify trailing-edge bond/serration-line continuity at 40–80% at higher zoom.
4. Clean stained/streaked zones before any repair decision to separate contamination from coating wear.

Maintain periodic drone inspection for trend tracking; re-image tip next cycle to monitor erosion

5.1.2 Suction Side - Segment-by-Segment Findings

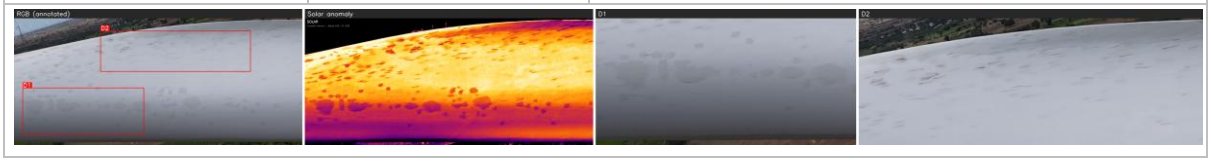
Suction Side (SS)	
	

SS Segment — 0% (Blade Root Hub)		3 finding(s) | max severity: Low
Annotated View	*Anomaly Visualization*	*Defect Detail Evidence*



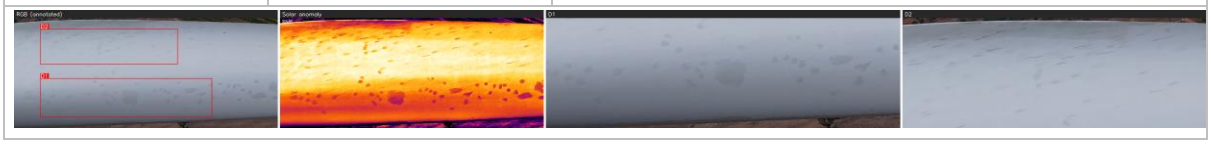
The suction-side root barrel shows widespread blotchy gel-coat mottling / patchy discoloration and numerous scattered small dark linear / dash marks across the surface – most consistent with weathering, coating-patch unevenness and surface soiling. The linear marks should be verified close-up to rule out hairline gel-coat cracking. No structural cracks, laminate exposure or edge damage are confirmed; the root collar is intact in view.

SS Segment — 20% (Root to Mid-Span)		2 finding(s) | max severity: Low
Annotated View	*Anomaly Visualization*	*Defect Detail Evidence*



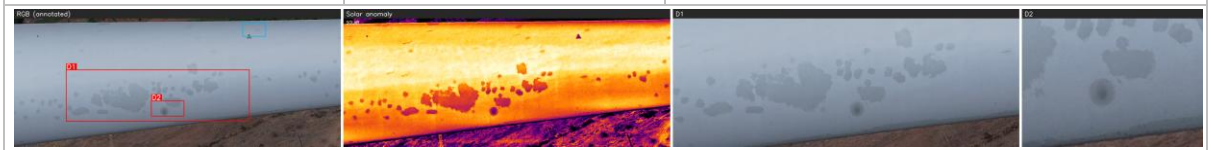
Uniform SS panel with a blotchy gel-coat mottling band (lower) and scattered dark dash marks (upper). Indications are consistent with weathering / soiling; no structural indications observed.

SS Segment — 40% (Mid-Span)		2 finding(s) | max severity: Low
Annotated View	*Anomaly Visualization*	*Defect Detail Evidence*



Midspan SS panel with grey gel-coat mottling patches (lower band) and scattered short dark marks. Surface-condition only; no crack-like propagation evident.

SS Segment — 60% (Mid-Span)		2 finding(s) | max severity: Low
Annotated View	*Anomaly Visualization*	*Defect Detail Evidence*



Markers: M1: triangular zone / position marker – existing label, not a defect.

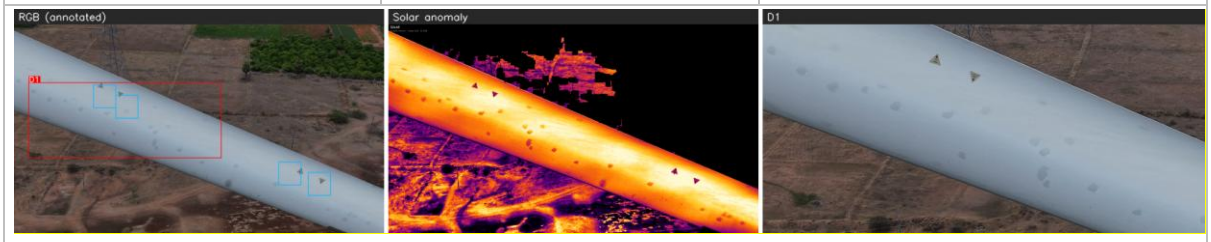
SS panel with a more pronounced darker smudge / blotch band (centre-lower) and a small dark spot; a triangular zone marker (existing label) is present and boxed in cyan. Blotches are consistent with soiling / coating discoloration; the small spot to be verified. No structural damage evident.

SS Segment — 70% (Mid-Span)		2 finding(s) | max severity: Low
Annotated View	*Anomaly Visualization*	*Defect Detail Evidence*



SS panel with scattered blotch / mottling patches and dark marks near the lower edge. Surface-condition only; no structural indications.

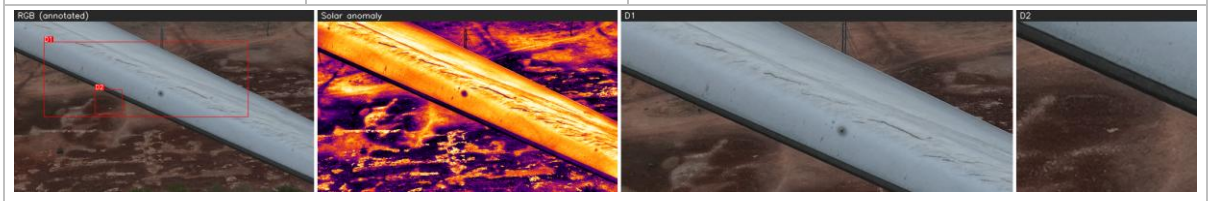
SS Segment — 80% (Mid-Span to Tip)		1 finding(s) | max severity: Low
Annotated View	*Anomaly Visualization*	*Defect Detail Evidence*



Markers: M1–M4: triangular zone / position markers – existing labels, not defects.

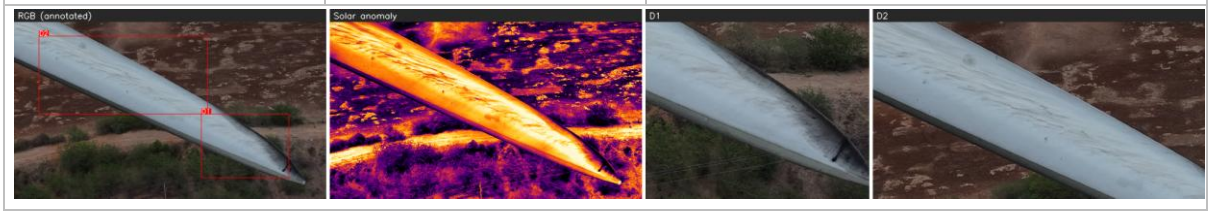
Outboard SS panel is comparatively clean with only faint surface blotches / soiling. Four triangular zone markers (existing labels, boxed in cyan) are present. No cracks or edge damage are evident.

SS Segment — 90% (Blade Tip)		2 finding(s) | max severity: Low
Annotated View	*Anomaly Visualization*	*Defect Detail Evidence*



Outboard SS shows longitudinal streaking / surface weathering consistent with erosion-related wear along the outboard leading edge, plus a small dark spot / mark (possible lightning receptor). No open crack is confirmed; verify at higher zoom.

SS Segment — 100% (Blade Tip)		2 finding(s) | max severity: Medium
Annotated View	*Anomaly Visualization*	*Defect Detail Evidence*



Tip region – the most significant SS finding: darkening at the leading-edge tip consistent with leading-edge erosion / coating wear, with longitudinal streaking along the outboard. The tip / edge is a high-stress, high-risk zone and warrants close-up verification. The trailing edge appears continuous.

Overall Findings – Consolidated SS Defects

Span	ID	Observed Condition	Category	Severity	Recommendation
0%	D1	Broad blotchy gel-coat mottling / patchy discoloration across the mid-barrel	Gel-coat weathering / coating-patch unevenness / soiling	**Low**	Clean and re-image; monitor at next cycle.
	D2	Numerous scattered dark linear / dash marks across the surface	Surface soiling streaks / minor gel-coat marks (rule out hairline cracking)	**Low**	Close-up verification to confirm surface-only; clean and re-image.
	D3	Darker smudge / soiling cluster right of centre	Localised soiling / staining	**Low**	Clean and monitor.
20%	D1	Blotchy gel-coat mottling band on the lower panel	Gel-coat weathering / soiling	**Low**	Clean and re-image; monitor.
	D2	Scattered dark dash / linear marks on the upper panel	Surface soiling streaks (rule out hairline cracking)	**Low**	Close-up verification; clean.
40%	D1	Grey gel-coat mottling patches on the lower band	Gel-coat unevenness / soiling	**Low**	Clean and re-image.
	D2	Scattered short dark marks across the panel	Surface soiling / minor marks (verify)	**Low**	Close-up verification; clean.

Span	ID	Observed Condition	Category	Severity	Recommendation
60%	D1	Darker smudge / blotch band (centre-lower)	Surface soiling / coating discoloration	**Low**	Clean and re-image; monitor.
	D2	Small dark spot / mark on the panel	Localised mark / soiling (verify)	**Low**	Close-up verification; clean.
70%	D1	Blotch / mottling patches across the panel	Gel-coat unevenness / soiling	**Low**	Clean and re-image.
	D2	Dark marks / spots near the lower edge	Surface soiling / minor marks	**Low**	Clean; monitor.
80%	D1	Faint surface blotches / soiling on the outboard panel	General soiling / gel-coat unevenness	**Low**	Monitor and clean at next cycle.
90%	D1	Longitudinal streaking / surface weathering along the outboard	Erosion-related surface wear / weathering streaks	**Low**	Clean and re-image; monitor progression.
	D2	Small dark spot / mark mid-blade	Localised mark / possible lightning receptor	**Low**	Close-up verification; confirm receptor condition.
100%	D1	Darkened band at the leading-edge tip	Leading-edge erosion / tip coating wear	**Medium**	Close-up inspection to confirm erosion depth; schedule LE repair / recoat and LE protection if gel-coat / laminate exposure is confirmed.
	D2	Longitudinal streaking / surface weathering along the outboard LE	Erosion-related surface wear / weathering	**Low**	Clean and re-image; monitor progression.

Overall Findings Summary

A total of 16 indications were identified across the eight suction-side span stations (0 Critical, 0 High, 1 Medium, 15 Low). The suction side is dominated by cosmetic surface-condition findings – widespread blotchy gel-coat mottling / patchy discoloration and scattered dark linear / dash marks across the inboard and midspan panels (0–70%), with faint outboard soiling. The single Medium-severity item is leading-edge erosion at the tip (100% D1).

The scattered dark linear / dash marks on the inboard SS are most consistent with surface soiling and minor gel-coat marks; a close-up verification pass is recommended to positively rule out hairline gel-coat cracking. No confirmed structural cracks, delamination, open edges or laminate exposure were observed on the suction side in the captured imagery.

Conclusion

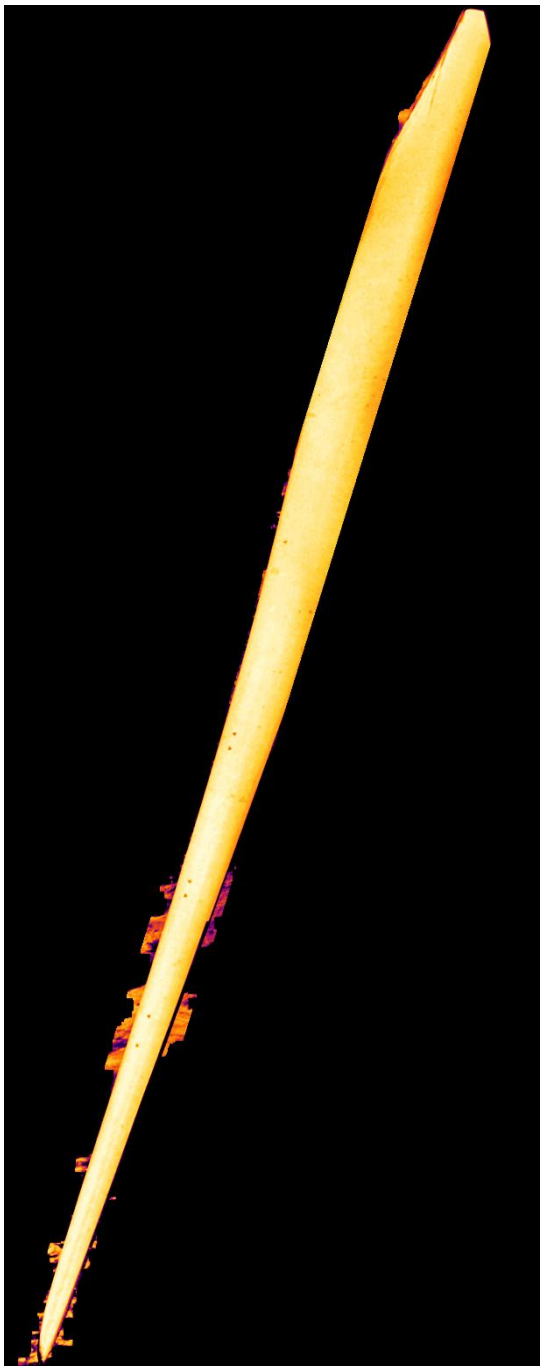
- The suction side of blade 5291 is serviceable: no confirmed cracks, delamination, open edges or bond-line separation in the captured imagery.
- Findings are predominantly cosmetic – gel-coat mottling / patchy discoloration, soiling and scattered surface marks across the inboard and midspan, with light outboard soiling.
- The most significant item is Medium-severity leading-edge erosion at the tip (100% D1). Continued operation is acceptable subject to the actions below.

Recommendations

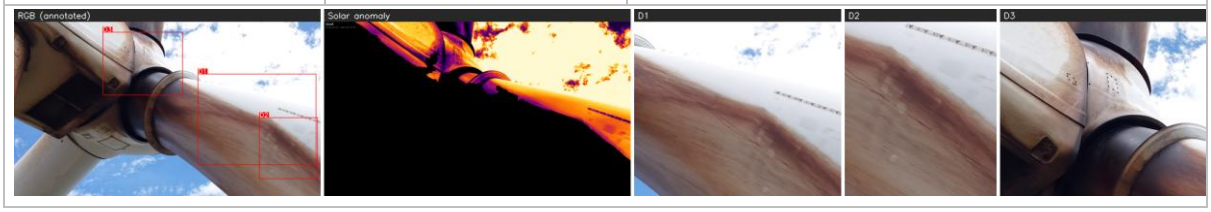
1. Priority close-up of the tip leading-edge erosion (100% D1); schedule LE repair / recoat + LE protection if gel-coat / laminate exposure is confirmed.
2. Close-up verification pass over the inboard SS linear / dash marks (0–40%) to positively rule out hairline gel-coat cracking.
3. Clean-and-reinspect the mottled / soiled panels to separate soiling from true coating wear before any repair decision.
4. Confirm the condition of the possible lightning receptor at the 90% station.
5. Maintain a standardized periodic drone inspection for consistent trend tracking; re-image the tip next cycle to monitor erosion progression.

5.2 Blade 5297

5.2.1 Pressure Side - Segment-by-Segment Findings

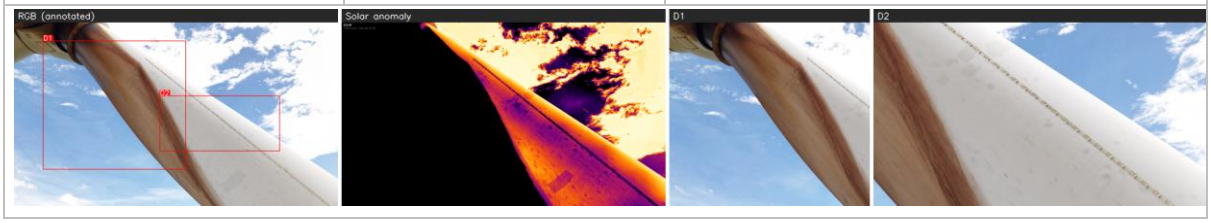
Pressure Side (PS)	
	

PS Segment — 0% (Blade Root Hub)		3 finding(s) | max severity: Medium
Annotated View	*Anomaly Visualization*	*Defect Detail Evidence*



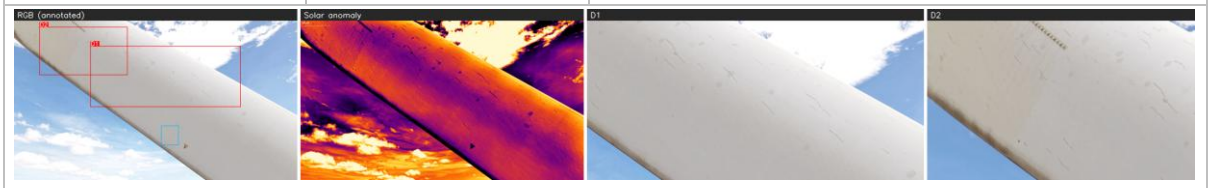
The root / hub region shows heavy rust and brown runoff staining across the hub casting, pitch-bearing collar and the blade-root barrel, together with longitudinal dark crack-like line(s) on the root barrel. The staining is consistent with weathering and runoff, but the crack-like line(s) warrant close-up verification / NDT to rule out gel-coat or structural cracking. No open structural failure is confirmed from the imagery.

PS Segment — 20% (Root to Mid-Span)		2 finding(s) | max severity: Medium
Annotated View	*Anomaly Visualization*	*Defect Detail Evidence*



A prominent brown / rust staining band runs along the leading-edge side of the inboard panel, with an associated dark line within the stain; the trailing-edge bond / serration line is visible and appears continuous. The staining is consistent with runoff, but the dark line should be checked close-up to rule out an underlying gel-coat crack.

PS Segment — 40% (Mid-Span)		2 finding(s) | max severity: Low
Annotated View	*Anomaly Visualization*	*Defect Detail Evidence*



Markers: M1: triangular zone / position marker – existing label, not a defect.

Midspan PS panel is largely clean with scattered dark linear marks / scratches and faint soiling spots. A triangular zone marker (existing label, boxed in cyan) is present. Marks appear superficial; a close-up pass is recommended to confirm they are surface-only.

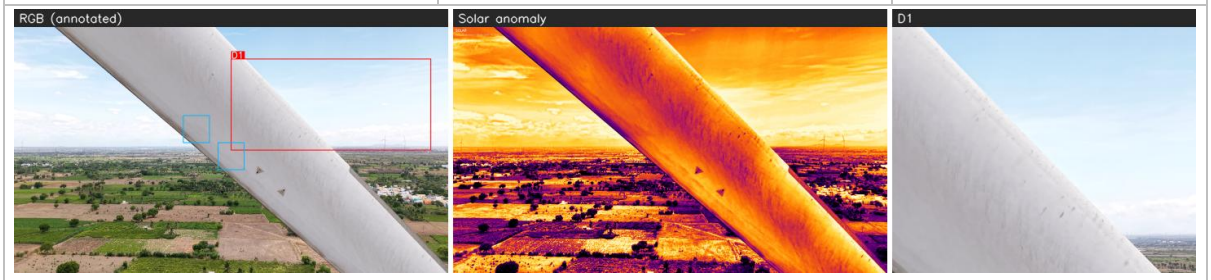
PS Segment — 60% (Mid-Span)		2 finding(s) | max severity: Low
Annotated View	*Anomaly Visualization*	*Defect Detail Evidence*



Markers: M1: triangular zone / position marker – existing label, not a defect.

PS panel with a localised grey scuff / smudge mark and faint surface spots. A triangular zone marker (existing label, boxed in cyan) is present. Surface-condition only; no structural indications observed.

PS Segment — 80% (Mid-Span to Tip)		1 finding(s) | max severity: Low
Annotated View	*Anomaly Visualization*	*Defect Detail Evidence*



Markers: M1–M2: triangular zone / position markers – existing labels, not defects.

Outboard PS panel shows trailing-edge streaking / soiling with faint grey mottling. Two triangular zone markers (existing labels, boxed in cyan) are present. No cracks or edge damage evident; the streaking is early erosion-related soiling to monitor.

PS Segment — 90% (Mid-Span to Tip)		2 finding(s) | max severity: Medium
Annotated View	*Anomaly Visualization*	*Defect Detail Evidence*



Markers: M1–M3: triangular zone / position markers — existing labels, not defects.

Outboard tip region shows leading-edge erosion with extensive grey / black surface streaking radiating from the leading edge, plus a discrete dark impact / erosion spot on the panel. This is an active erosion zone; close-up assessment is required to grade coating loss and rule out laminate exposure.

PS Segment — 100% (Blade Tip)		2 finding(s) | max severity: High
Annotated View	*Anomaly Visualization*	*Defect Detail Evidence*



Tip region – the most significant finding on this blade: advanced leading-edge erosion running to the tip, with a dark, rough, extensively streaked leading edge and darkened tip suggesting coating loss and possible laminate exposure. The tip is a high-stress, high-risk zone; this indication should be repaired on a priority basis and confirmed by close-up / NDT.

Overall Findings – Consolidated PS Defects

Span	ID	Observed Condition	Category	Severity	Recommendation
0%	D1	Heavy rust / brown runoff staining across the hub, pitch-bearing collar and blade-root barrel	Runoff / weathering staining (hub / pitch-bearing)	**Medium**	Clean and re-image; trace runoff source at the hub / pitch-bearing seals; treat corrosion staining.
	D2	Longitudinal dark crack-like line(s) on the blade-root barrel	Possible gel-coat / surface cracking (verify)	**Medium**	Priority close-up inspection to confirm crack depth; NDT / rope-access if a crack is confirmed.
	D3	Root-collar / pitch-bearing casting weathering and staining	Weathering / surface corrosion	**Low**	Monitor; clean at next access.
20%	D1	Brown / rust staining band along the leading-edge side with an associated dark line	Runoff staining; rule out underlying gel-coat crack	**Medium**	Clean and re-image; close-up to rule out a crack beneath the stain.
	D2	Trailing-edge bond / serration line	TE bond line (verification)	**Low**	Verify TE continuity at higher zoom.
40%	D1	Scattered dark linear marks / scratches across the panel	Surface scratches / minor marks (verify)	**Low**	Close-up verification to confirm surface-only; clean.

Span	ID	Observed Condition	Category	Severity	Recommendation
	D2	Faint soiling / spots on the panel	General soiling	**Low**	Clean; monitor.
60%	D1	Grey scuff / smudge mark on the panel	Surface scuff / soiling	**Low**	Clean and re-image.
	D2	Faint surface spots / marks	Minor soiling	**Low**	Clean; monitor.
80%	D1	Trailing-edge streaking / soiling along the outboard	TE soiling / early erosion-related wear	**Low**	Clean; verify TE at higher zoom; monitor progression.
90%	D1	Leading-edge erosion with extensive surface streaking radiating from the LE	Leading-edge erosion / coating wear	**Medium**	Close-up to confirm erosion depth; schedule LE repair / recoat + LE protection.
	D2	Discrete dark impact / erosion spot on the panel	Localised erosion / impact mark	**Medium**	Close-up inspection; repair if coating / laminate is compromised.
100%	D1	Advanced leading-edge erosion to the tip (dark, rough, extensively streaked), possible laminate exposure	Leading-edge erosion / tip coating loss	**High**	Repair within 3 months – LE repair / recoat + LE protection; confirm laminate condition by close-up / NDT; monitor until repaired.
	D2	Surface erosion streaking along the outboard toward the tip	Erosion-related surface wear	**Medium**	Clean and re-image; monitor progression as part of the LE repair scope.

Overall Findings Summary

A total of 14 indications were identified across the 7 pressure-side span stations (0 Critical, 1 High, 6 Medium, 7 Low). This is the most degraded side inspected in the campaign. The dominant themes are (1) heavy rust / brown runoff staining with longitudinal crack-like line(s) at the root / hub (0–20%), and (2) progressing leading-edge erosion from the outboard to the tip. The single High-severity item is advanced leading-edge erosion at the blade tip (100% D1), with associated Medium erosion at 90% and Medium root staining / crack-like indications at the root.

The root crack-like line(s) (0% D2) and the tip leading-edge erosion (100% D1) require priority close-up / NDT to confirm depth and laminate condition. Midspan findings are cosmetic surface marks and soiling. No open structural failure or bond-line separation was confirmed from the imagery, but the tip erosion and root indications mean the blade is serviceable only with a defined repair action.

Conclusion

- The pressure side of blade 5297 is serviceable but requires a defined repair action: no open structural failure or bond-line separation was confirmed, but the tip leading-edge erosion is advanced.
- The most significant item is High-severity leading-edge erosion at the tip (100% D1) – schedule repair within 3 months. Root rust staining with crack-like line(s) (0%) and 90% LE erosion are Medium items to verify close-up.
- Continued operation is acceptable in the short term provided the tip leading-edge erosion is repaired within ~3 months and the root crack-like indication is verified by close-up / NDT; monitor both until closed out.

Recommendations

1. Priority: repair the tip leading-edge erosion (100% D1) within ~3 months – LE repair / recoat + LE protection; confirm laminate condition by close-up / NDT.
2. Verify the root crack-like line(s) (0% D2) by close-up / NDT; clean and treat the root / hub rust runoff staining and trace its source at the pitch-bearing seals.
3. Clean-and-reinspect the mottled / soiled panels to separate soiling from true coating wear before any repair decision.
4. Confirm the condition of the possible lightning receptor at the 90% station.
5. Maintain a standardized periodic drone inspection for consistent trend tracking; re-image the tip next cycle to monitor erosion progression.

5.2.2 Suction Side - Segment-by-Segment Findings

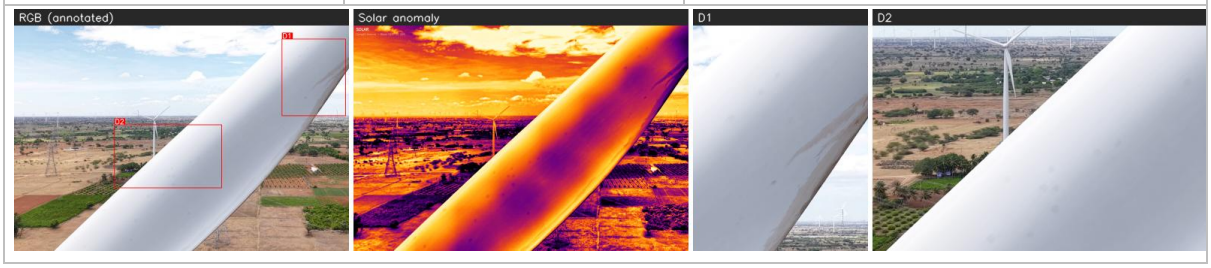
Suction Side (SS)	
	

SS Segment — 0% (Blade Root Hub)		2 finding(s) | max severity: Low
Annotated View	*Anomaly Visualization*	*Defect Detail Evidence*



The suction-side root barrel is in good condition – predominantly clean white gel-coat. Rust / brown staining is present at the root-collar / hub interface (runoff from the hub / spinner and tower), with only light scuff / scratch marks and faint streaks on the barrel itself. No structural cracks, laminate exposure or edge damage are evident.

SS Segment — 20% (Root to Mid-Span)		2 finding(s) | max severity: Low
Annotated View	*Anomaly Visualization*	*Defect Detail Evidence*



Clean SS panel. A tan / brown stain streak is present toward the root-ward edge; the remainder shows only faint surface marks / light soiling. No structural indications observed.

SS Segment — 40% (Mid-Span)		1 finding(s) | max severity: Low
Annotated View	*Anomaly Visualization*	*Defect Detail Evidence*



Markers: M1: triangular zone / position marker – existing label, not a defect.

Clean midspan SS panel with only faint scuff marks / light soiling. A triangular zone marker (existing label, boxed in cyan) is present. No crack-like features observed.

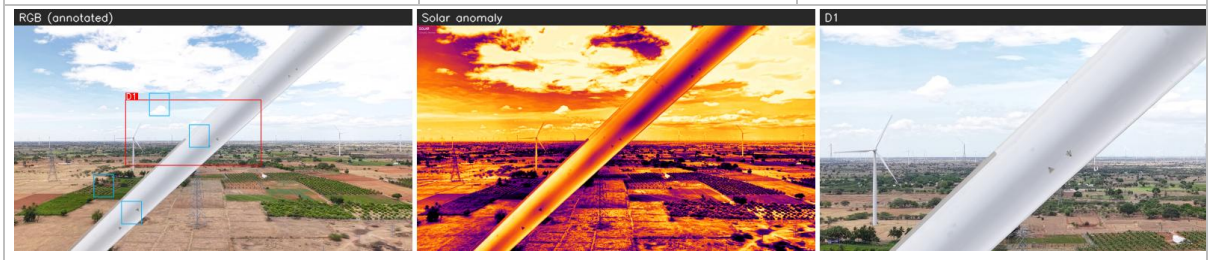
SS Segment — 60% (Mid-Span)		1 finding(s) | max severity: Low
Annotated View	*Anomaly Visualization*	*Defect Detail Evidence*



Markers: M1–M3: triangular zone / position markers – existing labels, not defects.

Clean SS panel with only faint surface spots / marks. Several triangular zone markers (existing labels, boxed in cyan) are present. No structural indications observed.

SS Segment — 80% (Mid-Span)		1 finding(s) | max severity: Low
Annotated View	*Anomaly Visualization*	*Defect Detail Evidence*



Markers: M1–M4: triangular zone / position markers – existing labels, not defects.

Outboard SS panel is clean with only faint surface marks / soiling. Multiple triangular zone markers (existing labels, boxed in cyan) are present. No cracks or edge damage are evident.

SS Segment — 90% (Mid-Span to Tip)		1 finding(s) | max severity: Low
Annotated View	*Anomaly Visualization*	*Defect Detail Evidence*



Markers: M1–M2: triangular zone / position markers – existing labels, not defects.

Outboard SS near the tip shows trailing-edge roughness / serration and faint scuffs; no open crack is confirmed. A triangular zone marker (existing label, boxed in cyan) is present. Verify trailing-edge condition at higher zoom.

SS Segment — 100% (Blade Tip)		2 finding(s) | max severity: Medium
Annotated View	*Anomaly Visualization*	*Defect Detail Evidence*



Tip region – the most significant SS finding: darkening / blackening at the leading-edge tip consistent with leading-edge erosion / coating wear, with trailing-edge serration and light streaking toward the tip. The tip / edge is a high-stress zone and warrants close-up verification.

Overall Findings – Consolidated SS Defects

Span	ID	Observed Condition	Category	Severity	Recommendation
0%	D1	Rust / brown staining at the root-collar / hub interface	Runoff staining from hub / spinner / tower	**Low**	Clean and re-image; trace runoff source at the hub / collar; monitor.
	D2	Light scuff / scratch marks and faint streaks on the root barrel	Minor handling / surface marks	**Low**	Clean and monitor.
20%	D1	Tan / brown stain streak toward the root-ward edge	Runoff / surface stain	**Low**	Clean and re-image; monitor.
	D2	Faint surface marks / light soiling on the panel	General soiling	**Low**	Clean; monitor.
40%	D1	Faint scuff marks / light soiling on the panel	Minor surface marks / soiling	**Low**	Clean and re-image.
60%	D1	Faint surface spots / marks on the panel	Minor soiling / marks	**Low**	Clean; monitor.
80%	D1	Faint surface marks / soiling on the outboard panel	General soiling / minor marks	**Low**	Monitor and clean at next cycle.
90%	D1	Trailing-edge roughness / serration and faint scuffs on the outboard	TE serration / minor surface wear	**Low**	Verify TE continuity at higher zoom; clean.

Span	ID	Observed Condition	Category	Severity	Recommendation
100%	D1	Darkened / blackened band at the leading-edge tip	Leading-edge erosion / tip coating wear	**Medium**	Close-up inspection to confirm erosion; schedule LE repair/recoat if gel-coat exposure is confirmed.
	D2	Trailing-edge serration / roughness and light streaking toward the tip	TE serration / erosion-related surface wear	**Low**	Verify TE continuity at higher zoom; clean and monitor.

Overall Findings Summary

A total of 10 indications were identified across the 7 suction-side span stations (0 Critical, 0 High, 1 Medium, 9 Low). Blade 5297 SS is in very good condition – predominantly clean white gel-coat with only light scuff / scratch marks and faint soiling, plus rust / brown runoff staining at the root-collar / hub interface and a root-ward stain streak. The single Medium-severity item is leading-edge erosion at the tip (100% D1).

Trailing-edge roughness / serration at the 90–100% stations should be verified at higher zoom. No confirmed structural cracks, delamination, open edges or laminate exposure were observed on the suction side in the captured imagery. This blade is markedly cleaner than blade 5291 SS.

Conclusion

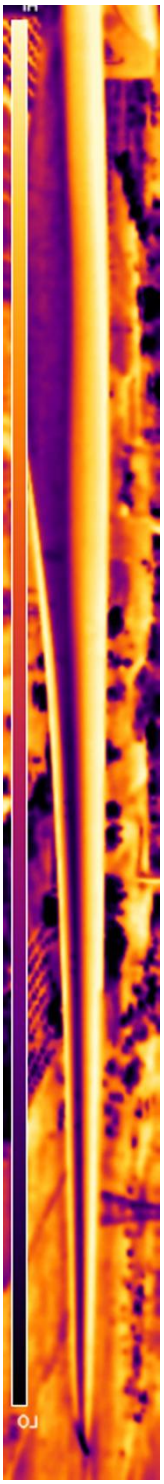
- The suction side of blade 5297 is serviceable: no confirmed cracks, delamination, open edges or bond-line separation in the captured imagery.
- Findings are limited and predominantly cosmetic – light scuff marks and soiling, root-collar / hub rust runoff staining, and a root-ward stain streak.
- The most significant item is Medium-severity leading-edge erosion at the tip (100% D1). Continued operation is acceptable subject to the actions below.

Recommendations

1. Priority close-up of the tip leading-edge erosion (100% D1); schedule LE repair / recoat + LE protection if gel-coat / laminate exposure is confirmed.
2. Trace and clean the root-collar / hub rust runoff staining; verify trailing-edge condition at the 90–100% stations at higher zoom.
3. Clean-and-reinspect the mottled / soiled panels to separate soiling from true coating wear before any repair decision.
4. Confirm the condition of the possible lightning receptor at the 90% station.
5. Maintain a standardized periodic drone inspection for consistent trend tracking; re-image the tip next cycle to monitor erosion progression.

5.3 Blade 5287

5.3.1 Pressure Side - Segment-by-Segment Findings

Pressure Side (PS)	
	

PS Segment — 0% (Blade Root Hub)		3 finding(s) | max severity: Medium
Annotated View	*Anomaly Visualization*	*Defect Detail Evidence*



The pressure-side root barrel shows the most significant findings of the survey. Heavy rust / brown runoff staining spreads across the lower root barrel from the root-collar / hub interface. An exposed / frayed trailing-edge band with serration-like roughness runs along the upper edge, and faint grey mottling / light soiling is present across the mid barrel. No open through-cracks are evident in the captured imagery.

PS Segment — 20% (Root to Mid-Span)		3 finding(s) | max severity: Low
Annotated View	*Anomaly Visualization*	*Defect Detail Evidence*



Clean PS panel with widespread grey mottling and dense spotting. A trailing-edge serration band with roughness runs along the upper edge, and faint tan / brown runoff staining is present toward the lower root-ward corner. No structural indications observed.

PS Segment — 40% (Mid-Span)		3 finding(s) | max severity: Low
Annotated View	*Anomaly Visualization*	*Defect Detail Evidence*



Midspan PS panel with extensive blotchy grey patch-like marks and mottling (possible prior repairs / coating unevenness). A trailing-edge bond-line band with light serration roughness runs along the mid seam, with faint tan / brown discolouration at the outboard-upper corner. No confirmed crack-like features observed.

PS Segment — 60% (Mid-Span)		2 finding(s) | max severity: Low
Annotated View	*Anomaly Visualization*	*Defect Detail Evidence*



PS panel with vertical streaking, blotchy grey patch-like marks and mottling. A trailing-edge bond-line band with light serration roughness runs along the seam. No structural indications observed.

PS Segment — 80% (Mid-Span to Tip)		2 finding(s) | max severity: Low
Annotated View	*Anomaly Visualization*	*Defect Detail Evidence*



Markers: M1: triangular zone / position markers — existing labels, not defects.

Outboard PS panel with diagonal dark streaks / smears and light soiling. A trailing-edge band with darkened staining / roughness runs along the upper edge. Triangular zone markers (existing labels, boxed in green) are present. No structural indications observed.

PS Segment — 100% (Blade Tip)		2 finding(s) | max severity: Medium
Annotated View	*Anomaly Visualization*	*Defect Detail Evidence*



Tip PS panel shows darkened / blackened blotchy staining and streaking at the leading-edge tip and a trailing-edge band with darkened staining / roughness toward the tip. Grey mottling, vertical streaking and light soiling are present along the outboard panel. Leading- and trailing-edge condition must be verified at higher zoom to confirm erosion versus staining.

Overall Findings — Consolidated PS Defects

Span	ID	Observed Condition	Category	Severity	Recommendation
0%	D1	Heavy rust / brown runoff staining across the lower root barrel from the root-collar / hub interface	Runoff staining from hub / spinner (weathering)	Medium	Trace runoff source at hub / collar; clean and re-image; confirm no gel-coat breach beneath staining; monitor.
	D2	Exposed / frayed trailing-edge band with serration-like roughness along the upper edge	Trailing-edge erosion / bond-line roughness	Medium	Verify TE bond-line continuity and serration condition at higher zoom; schedule TE repair if laminate exposure confirmed.
	D3	Faint grey mottling / spotting and light soiling across the mid barrel	Surface soiling / mottling	Low	Clean and monitor.
20%	D1	Widespread grey mottling and dense spotting across the panel	Surface soiling / mottling	Low	Clean and re-image to separate soiling from coating wear; monitor.
	D2	Trailing-edge serration band with roughness along the upper edge	Trailing-edge serration / minor wear	Low	Verify TE continuity at higher zoom; clean.
	D3	Faint tan / brown runoff staining toward the lower root-ward corner	Runoff / surface stain	Low	Clean and monitor.
40%	D1	Extensive blotchy grey patch-like marks and mottling across the midspan panel (possible prior repairs / coating unevenness)	Possible prior repair patches / coating marks + soiling	Low	Verify against repair records; monitor patch edges; clean and re-image.
	D2	Trailing-edge bond-line band with light serration roughness along the mid seam	Trailing-edge serration / bond-line wear	Low	Verify TE bond-line continuity at higher zoom; monitor.

Span	ID	Observed Condition	Category	Severity	Recommendation
	D3	Faint tan / brown discolouration at the outboard-upper corner	Surface stain / soiling	**Low**	Clean and monitor.
60%	**D1**	Vertical streaking, blotchy grey patch-like marks and mottling across the panel	Surface soiling / mottling + possible coating marks	**Low**	Clean and re-image; verify patch-like marks against repair records; monitor.
	D2	Trailing-edge bond-line band with light serration roughness along the seam	Trailing-edge serration / bond-line wear	**Low**	Verify TE bond-line continuity at higher zoom; monitor.
80%	**D1**	Diagonal dark streaks / smears and light soiling along the outboard panel	Surface soiling / streaks	**Low**	Clean; monitor.
	D2	Trailing-edge band with darkened staining / roughness along the upper edge	Trailing-edge staining / minor wear	**Low**	Verify TE continuity at higher zoom; clean.
100%	**D1**	Darkened / blackened blotchy staining and streaking at the leading-edge tip	Leading-edge erosion / tip coating wear	**Medium**	Close-up to confirm erosion depth; schedule LE repair / recoat + LE protection if gel-coat / laminate exposure confirmed.
	D2	Trailing-edge band with darkened staining and roughness toward the tip	Trailing-edge erosion / staining	**Medium**	Verify TE bond-line and erosion at higher zoom; schedule TE repair if laminate exposure confirmed.
	D3	Grey mottling, vertical streaking and light soiling along the outboard panel	Surface soiling / mottling + streaks	**Low**	Clean and re-image; monitor.

Overall Findings Summary

A total of 16 indications were identified across the 6 pressure-side span stations (0 Critical, 0 High, 4 Medium, 12 Low). Blade 5287 PS is in serviceable condition — predominantly clean white gel-coat with widespread grey mottling / soiling and blotchy patch-like marks (possible prior repairs) across the mid-span panels.

The most significant findings are concentrated at the root, trailing edge and tip: heavy rust / brown runoff staining from the root-collar / hub interface and an exposed / frayed trailing-edge band at the root (both Medium), and darkened leading- and trailing-edge staining / erosion at the tip (Medium). Trailing-edge serration / bond-line roughness is present along most stations and should be verified at higher zoom. No confirmed structural through-cracks, delamination or laminate exposure were positively identified in the captured imagery.

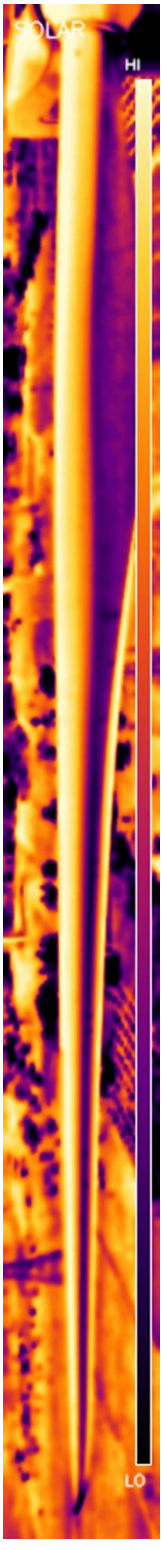
Conclusion

- The pressure side of blade 5287 is serviceable: no confirmed cracks, delamination, open edges or bond-line separation positively identified in the captured imagery.
- Findings are predominantly cosmetic — grey mottling / soiling and blotchy patch-like marks (possible prior repairs).
- Priority items are the root rust runoff and exposed trailing-edge band, and the tip leading- / trailing-edge staining / erosion — all Medium severity. Continued operation is acceptable subject to the actions below.

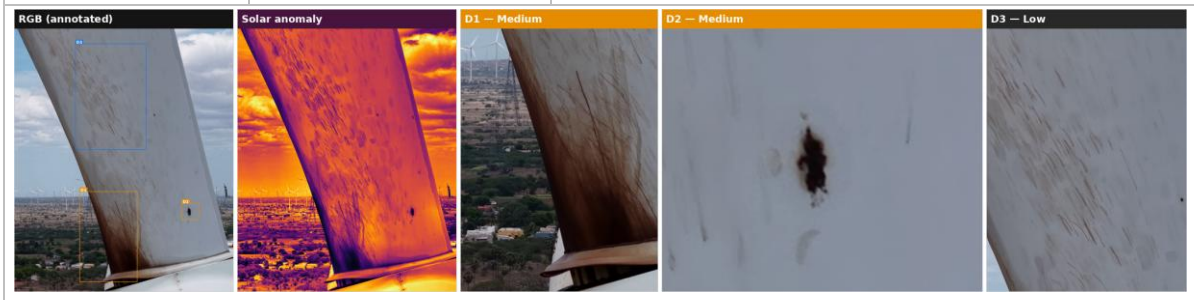
Recommendations

1. Trace and clean the root-collar / hub rust runoff staining (0% D1); confirm no gel-coat breach beneath the staining.
2. Priority close-up of the root trailing-edge exposed / frayed band (0% D2) and the tip trailing-edge band (100% D2); schedule TE repair if laminate exposure is confirmed.
3. Close-up of the tip leading-edge darkening (100% D1) to confirm erosion depth; schedule LE repair / recoat + LE protection if gel-coat / laminate exposure is confirmed.
4. Verify trailing-edge bond-line continuity and serration condition at all stations (20% / 40% / 60% / 80%) at higher zoom.
5. Clean-and-reinspect the mottled / patch-marked mid-span panels to separate soiling from true coating wear and confirm prior-repair status against records.
6. Maintain a standardized periodic drone inspection for consistent trend tracking; re-image the root, trailing edge and tip next cycle to monitor rust, TE and LE progression.

5.3.2 Suction Side - Segment-by-Segment Findings

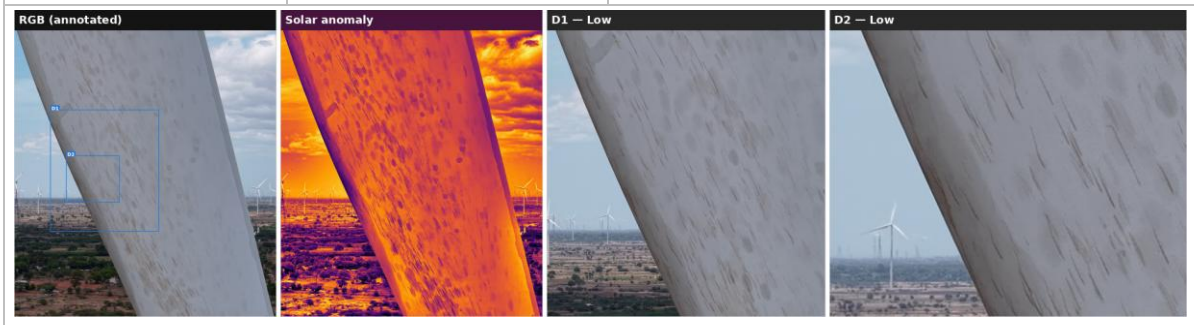
Suction Side (SS)	
	

SS Segment — 0% (Blade Root Hub)		3 finding(s) | max severity: Medium
Annotated View	*Anomaly Visualization*	*Defect Detail Evidence*



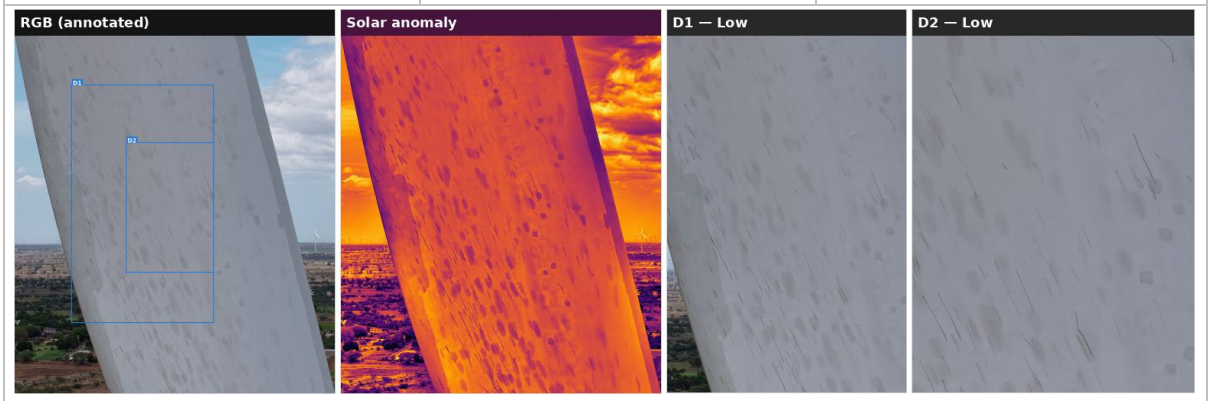
The suction-side root barrel is generally sound white gel-coat, but shows the most significant findings of the survey. Heavy rust / brown runoff staining streams down from the root-collar / hub interface across the lower barrel, and a distinct rust spot / possible pinhole-pit is present on the mid barrel. Diagonal scuff / scratch streaks and light soiling are present on the upper barrel. No open cracks, laminate exposure or edge damage are evident.

SS Segment — 20% (Root to Mid-Span)		2 finding(s) | max severity: Low
Annotated View	*Anomaly Visualization*	*Defect Detail Evidence*



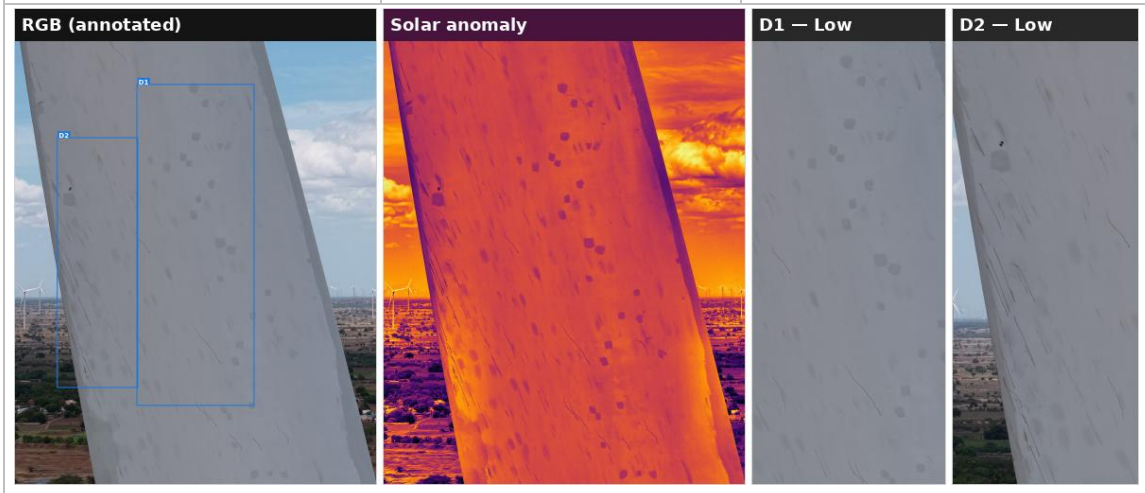
Clean SS panel with widespread light mottling, tan / brown spotting and diagonal scuff streaks. Faint greyish square patch-like marks (possible previous repair / filler patches) are present. No structural indications observed.

SS Segment — 40% (Mid-Span)		2 finding(s) | max severity: Low
Annotated View	*Anomaly Visualization*	*Defect Detail Evidence*



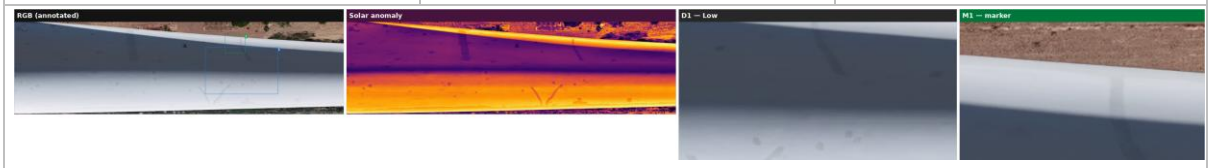
Midspan SS panel with diffuse grey blotches, tan streaks and scattered scuff marks. Faint thin dark curvilinear marks are present and should be verified as superficial. No confirmed crack-like features observed.

SS Segment — 60% (Mid-Span)		2 finding(s) | max severity: Low
Annotated View	*Anomaly Visualization*	*Defect Detail Evidence*



SS panel with a column of faint grey square patch-like marks and light mottling, plus thin dark curvilinear / hairline surface marks and scuffs on the leading-ward side. No structural indications observed.

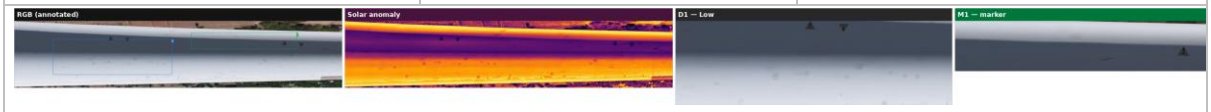
SS Segment — 70% (Mid-Span)		2 finding(s) | max severity: Low
Annotated View	*Anomaly Visualization*	*Defect Detail Evidence*



Markers: M1: triangular zone / position marker — existing label, not a defect.

Clean outboard SS panel with only faint irregular light streaks / smears. A triangular zone marker (existing label, boxed in green) is present. No cracks or edge damage evident.

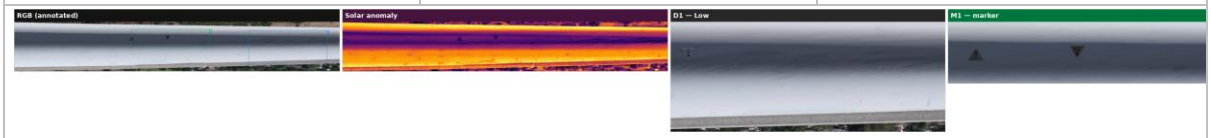
SS Segment — 80% (Mid-Span)		1 finding(s) | max severity: Low
Annotated View	*Anomaly Visualization*	*Defect Detail Evidence*



Markers: M1: triangular zone / position markers — existing labels, not defects.

Outboard SS panel is clean with only faint surface marks / soiling. Multiple triangular zone markers (existing labels, boxed in cyan) are present. No cracks or edge damage are evident.

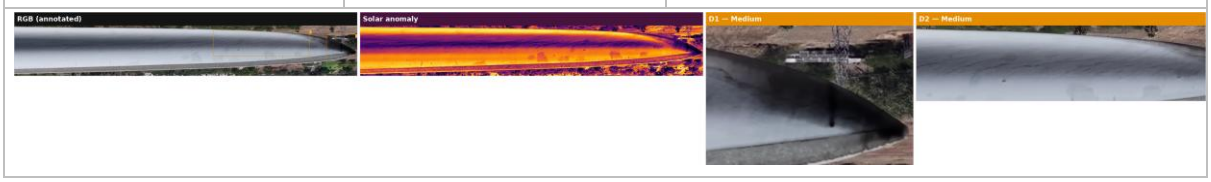
SS Segment — 90% (Mid-Span to Tip)		1 finding(s) | max severity: Low
Annotated View	*Anomaly Visualization*	*Defect Detail Evidence*



Markers: M1: triangular zone / position markers — existing labels, not defects.

Outboard SS panel with faint diagonal scuff / streak marks toward the tip. Triangular zone markers (existing labels, boxed in green) are present. Trailing-edge condition should be verified at higher zoom.

SS Segment — 100% (Blade Tip)		2 finding(s) | max severity: Medium
Annotated View	*Anomaly Visualization*	*Defect Detail Evidence*



Tip SS panel shows a darkened / blackened band and streaking at the leading-edge tip (erosion / tip coating wear) and fine dark hairline / thread-like marks running along the tip panel that must be verified as surface streaks and not hairline gel-coat cracks.

Overall Findings — Consolidated SS Defects

Span	ID	Observed Condition	Category	Severity	Recommendation
0%	D1	Heavy rust / brown runoff staining down the barrel from the root-collar / hub interface	Runoff staining from hub / spinner (weathering)	Medium	Trace runoff source at hub / collar; clean and re-image. Confirm no gel-coat breach beneath the staining; monitor.
	D2	Distinct rust spot / possible pinhole-pit with dark centre on the mid barrel	Localized corrosion stain / possible surface pit	Medium	Close-up inspection to rule out gel-coat pinhole / pit; clean and monitor; recoat if substrate is exposed.
	D3	Diagonal scuff / scratch streaks and light soiling across the upper barrel	Minor handling / surface marks + soiling	Low	Clean and monitor.
20%	D1	Widespread mottling, tan / brown spotting and diagonal scuff streaks across the panel	Surface soiling / mottling + minor scuffs	Low	Clean and re-image to separate soiling from coating wear; monitor.
	D2	Faint greyish square patch-like marks (possible previous repair / filler patches)	Possible prior repair patch / coating mark	Low	Verify against repair records; monitor for edge lifting.
40%	D1	Diffuse grey blotches, tan streaks and scattered scuff marks across the midspan panel	Surface soiling / mottling + minor scuffs	Low	Clean and re-image; monitor.
	D2	Faint hairline / thin dark curvilinear marks on the panel	Superficial surface marks (possible gel-coat scratch)	Low	Verify at higher zoom that marks are superficial and not gel-coat cracks; monitor.

Span	ID	Observed Condition	Category	Severity	Recommendation
60%	**D1**	Column of faint grey square patch-like marks and light mottling down the panel	Possible prior repair patches / coating marks + soiling	**Low**	Verify against repair records; monitor patch edges; clean.
	D2	Thin dark curvilinear / hairline surface marks and scuffs on the leading-ward side	Superficial surface marks	**Low**	Confirm superficial at higher zoom; clean and monitor.
70%	**D1**	Faint irregular light streaks / smears on the outboard panel	Light surface soiling / smears	**Low**	Clean; monitor.
80%	**D1**	Faint surface spots / light soiling on the outboard panel	General soiling / minor marks	**Low**	Clean at next cycle; monitor.
90%	**D1**	Faint diagonal scuff / streak marks along the outboard panel toward the tip	Minor surface wear / soiling	**Low**	Verify trailing-edge continuity at higher zoom; clean.
100%	**D1**	Darkened / blackened band and streaking at the leading-edge tip	Leading-edge erosion / tip coating wear	**Medium**	Close-up to confirm erosion depth; schedule LE repair / recoat + LE protection if gel-coat / laminate exposure is confirmed.
	D2	Fine dark hairline / thread-like marks and streaks running along the tip panel	Tip surface wear (verify not hairline cracks)	**Medium**	Verify at higher zoom these are surface streaks and not hairline gel-coat cracks; clean and monitor closely.

Overall Findings Summary

A total of 14 indications were identified across the 8 suction-side span stations (0 Critical, 0 High, 4 Medium, 10 Low). Blade 5287 SS is in serviceable condition — predominantly clean white gel-coat with light scuff / scratch marks, surface mottling / soiling and faint square patch-like marks (possible prior repairs) across the mid-span panels.

The most significant findings are at the root and tip: heavy rust / brown runoff staining from the root-collar / hub interface and a distinct rust spot / possible pinhole-pit on the root barrel (both Medium), and a darkened leading-edge band with fine hairline streaking at the tip (Medium) that must be verified as surface wear rather than hairline gel-coat cracks. No confirmed structural cracks, delamination, open edges or laminate exposure were observed in the captured imagery.

Conclusion

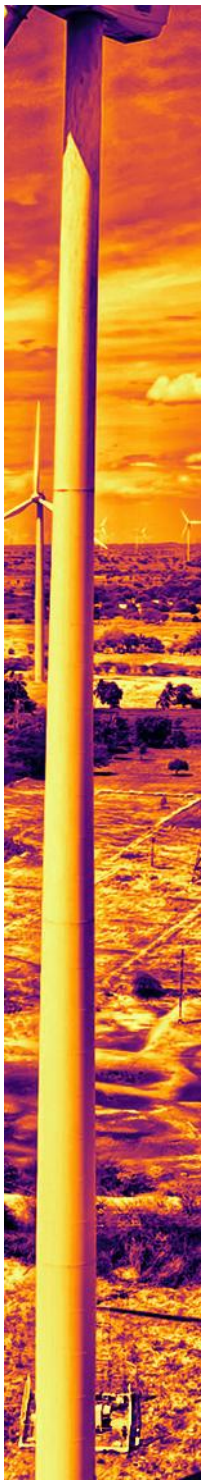
- The suction side of blade 5287 is serviceable: no confirmed cracks, delamination, open edges or bond-line separation in the captured imagery.
- Findings are predominantly cosmetic — light scuff marks, surface mottling / soiling and possible prior-repair patches.
- The priority items are the root rust runoff and rust spot / pit, and the tip leading-edge darkening / hairline streaking — all Medium severity. Continued operation is acceptable subject to the actions below.

Recommendations

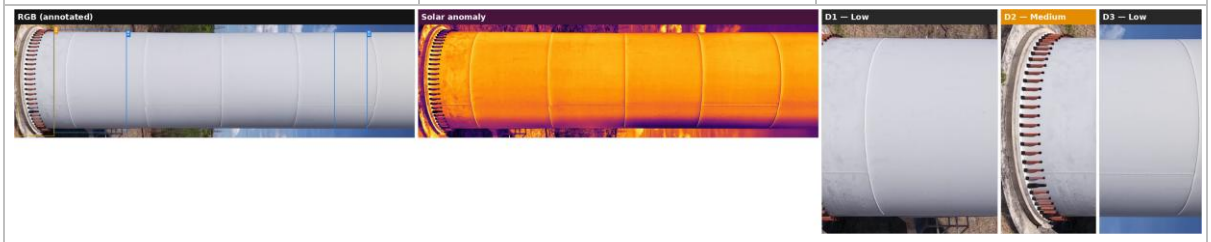
1. Priority close-up of the root rust spot / possible pinhole-pit (0% D2) to rule out gel-coat breach; recoat if substrate is exposed.
2. Trace and clean the root-collar / hub rust runoff staining (0% D1); confirm no gel-coat breach beneath the staining.
3. Close-up of the tip leading-edge darkening (100% D1) to confirm erosion depth; schedule LE repair / recoat + LE protection if gel-coat / laminate exposure is confirmed.
4. Verify the tip hairline / thread-like marks (100% D2) and the 40% / 60% curvilinear marks at higher zoom to separate superficial scratches from gel-coat cracks.
5. Clean-and-reinspect the mottled / soiled and patch-marked mid-span panels to separate soiling from true coating wear and confirm prior-repair status against records.
6. Verify trailing-edge continuity at the 90–100% stations at higher zoom.
7. Maintain a standardized periodic drone inspection for consistent trend tracking; re-image the root and tip next cycle to monitor the rust and LE erosion.

6. Tower Inspection

6.1 Structural Tower Pillar - Segment-by-Segment Findings

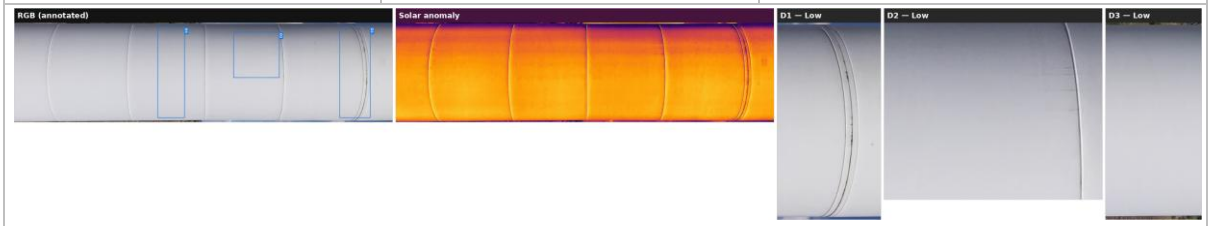
Structural Tower	
	

Tower Base | Flange / Anchor Ring (0%)		3 finding(s) | max severity: Medium
Annotated View	*Anomaly Visualization*	*Defect Detail Evidence*



The tower base barrel is in good condition — clean white coating with only light soiling / grey streaking and faint water-run marks above the base flange. The base anchor-bolt ring is exposed at the foundation interface and should be verified at close range for cap condition and corrosion. Circumferential can-to-can shell weld seams are intact with continuous paint.

Lower Tower | Shell Sections (~25%)		3 finding(s) | max severity: Low
Annotated View	*Anomaly Visualization*	*Defect Detail Evidence*



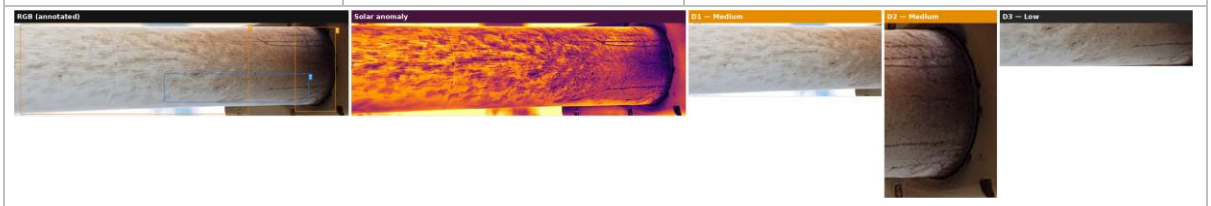
Lower tower shell sections are clean and sound with only dark scuff / contact marks and minor black streaking along the upper circumferential seam, and faint vertical drip / run streaks. Circumferential shell seams show continuous paint. No structural indications observed.

Mid Tower | Flange Connection (Upper) (~50%)		3 finding(s) | max severity: Low
Annotated View	*Anomaly Visualization*	*Defect Detail Evidence*



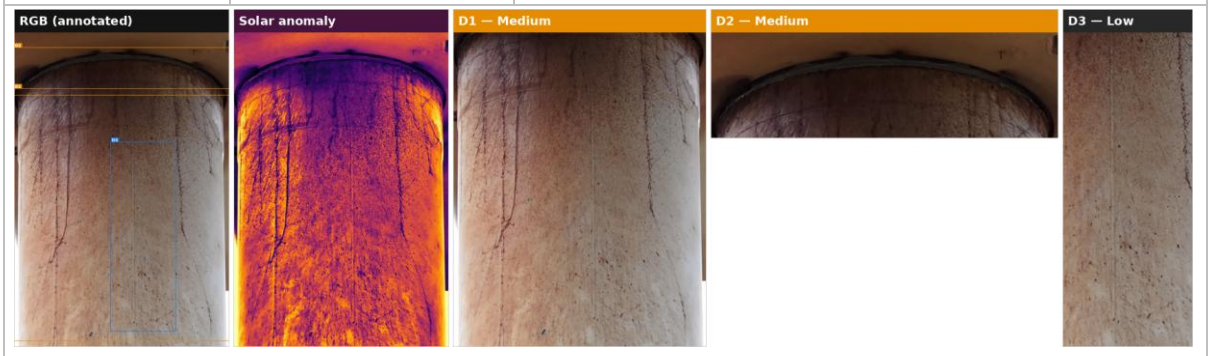
Lower mid-tower flange connection shows rust / brown staining, dark discolouration and paint disturbance along the bolted flange-connection ring, with a bright horizontal scuff / abrasion band and paint burnish immediately below the ring. Vertical brush / streak marks and light soiling are present across the shell. The flange joint should be inspected at close range for corrosion, bolt condition and sealant.

Upper Tower | Nacelle-Root Interface (~75%)		3 finding(s) | max severity: Medium
Annotated View	*Anomaly Visualization*	*Defect Detail Evidence*



The upper tower shows extensive rust / brown runoff staining streaming down from the nacelle-root / yaw-ring interface, with dark corrosion staining and paint disturbance at the top shell-to-nacelle joint and vertical rust rivulets running down the shell. The runoff source at the nacelle base / yaw seal should be traced and rectified; verify no coating breach beneath the staining.

Tower Top | Nacelle Base / Yaw Ring		3 finding(s) | max severity: Medium
Annotated View	*Anomaly Visualization*	*Defect Detail Evidence*



Markers: M1: triangular zone / position markers — existing labels, not defects.

The tower top shows heavy widespread rust / brown runoff staining and rust rivulets across the upper shell from the nacelle base / yaw-ring seal, with a dark corrosion band and paint disturbance along the nacelle-to-tower top joint / bolt ring. This is the most significant corrosion-staining zone of the survey. The runoff source must be traced and rectified and the top joint / bolt ring inspected at close range for active corrosion.

Overall Findings — Consolidated Tower Defects

Segment	ID	Observed Condition	Category	Severity	Recommendation
Base (0%)	**D1**	Light soiling / grey streaking and faint water-run marks on the lower barrel above the base flange	Surface soiling / water staining	**Low**	Clean and monitor.
	D2	Base anchor-bolt ring: bolt caps / covers exposed at the foundation interface	Anchor-bolt cap / foundation interface (inspect)	**Medium**	Verify anchor-bolt cap condition, corrosion and torque marks at close range; confirm grout / base seal integrity.
	D3	Horizontal circumferential weld / can-to-can shell seams	Shell circumferential weld seam (monitor)	**Low**	Monitor weld-seam paint continuity; log for trend.
Base +25%	**D1**	Dark scuff / contact marks and minor black streaking along the upper circumferential seam	Surface scuff / contact marks + soiling	**Low**	Clean and monitor.
	D2	Faint vertical drip / run streaks below the upper seam	Water-run staining	**Low**	Clean and monitor.
	D3	Circumferential shell seams — paint continuity intact	Shell circumferential seam (monitor)	**Low**	Monitor seam paint continuity; log for trend.
Connection L2	**D1**	Dark linear staining / debris accumulation along the circumferential flange-connection seam	Flange-seam soiling / staining	**Low**	Clean and re-image; verify seam sealant / paint continuity at higher zoom.
	D2	Faint grey vertical streaking / water-run marks down the shell	Water-run staining	**Low**	Clean and monitor.

Segment	ID	Observed Condition	Category	Severity	Recommendation
	D3	Lower circumferential connection seam with light contact scuffing	Shell connection seam (monitor)	**Low**	Monitor seam condition; log for trend.
Connection L1	**D1**	Rust / brown staining, dark discolouration and paint disturbance along the bolted flange-connection ring	Flange-ring corrosion staining / paint disturbance	**Medium**	Close-up inspection of the flange joint for corrosion, bolt condition and sealant; recoat / touch-up if paint breach or corrosion confirmed.
	D2	Bright horizontal scuff / abrasion band and paint burnish immediately below the flange ring	Surface abrasion / paint burnish	**Low**	Clean; verify no substrate exposure; touch-up coating if needed.
	D3	Vertical brush / streak marks and light soiling across the shell	Surface soiling / brush marks	**Low**	Clean and monitor.
Root +25%	**D1**	Extensive rust / brown runoff staining down the shell from the nacelle-root / yaw-ring interface	Runoff staining from nacelle / yaw interface (weathering)	**Medium**	Trace runoff source at the nacelle base / yaw seal; clean and re-image; confirm no coating breach beneath staining; monitor.
	D2	Dark corrosion staining and paint disturbance at the top shell-to-nacelle joint / ring	Top-joint corrosion staining / paint disturbance	**Medium**	Close-up of the shell-to-nacelle joint for corrosion and seal condition; recoat if breach confirmed.
	D3	Vertical rust rivulets / streak lines running down the shell	Rust rivulet staining	**Low**	Clean and monitor; address upstream source.

Segment	ID	Observed Condition	Category	Severity	Recommendation
Root (Top)	**D1**	Heavy widespread rust / brown runoff staining and rust rivulets across the upper shell from the nacelle base / yaw-ring seal	Severe runoff staining from nacelle / yaw seal (weathering)	**Medium**	Trace and rectify runoff source at the nacelle base / yaw seal; clean; confirm no coating breach or active corrosion beneath staining; recoat if required.
	D2	Dark corrosion band and paint disturbance along the nacelle-to-tower top joint / bolt ring	Top-joint corrosion staining / bolt-ring disturbance	**Medium**	Close-up of the top joint and bolt ring for corrosion, bolt and seal condition; recoat / repair if breach or active corrosion confirmed.
	D3	Faint vertical shell seam and scattered dark speckle staining on the shell	Shell seam + speckle staining	**Low**	Clean and monitor; log for trend.

Overall Findings Summary

A total of 18 indications were identified across the 6 tower segments (0 Critical, 0 High, 6 Medium, 12 Low). The tower / base pillar is in serviceable condition — the lower and mid barrel are predominantly clean white coating with light soiling, water-run streaking, contact scuffs and intact circumferential weld / flange seams.

Findings are concentrated at the interfaces: the base anchor-bolt ring and lower flange connection show corrosion staining and paint disturbance to be verified at close range, and the upper tower / tower-top nacelle-root interface shows extensive rust / brown runoff staining and rust rivulets from the nacelle base / yaw-ring seal (the most significant zone of the survey). No confirmed structural cracks, weld failures or section loss were positively identified in the captured imagery; the staining and paint disturbances require close-range and, where noted, NDT verification.

Conclusion

- The tower / base pillar is serviceable: no confirmed structural cracks, weld failures, flange separation or section loss positively identified in the captured imagery.
- The lower and mid tower barrel findings are predominantly cosmetic — soiling, water-run staining, contact scuffs and intact seams.
- Priority items are the nacelle-root / yaw-seal runoff staining at the tower top, the lower flange-ring corrosion staining, and the base anchor-bolt ring — all Medium severity pending close-range verification. Continued operation is acceptable subject to the actions below.

Recommendations

1. Trace and rectify the nacelle-root / yaw-ring seal runoff at the tower top (Root +25% & Top D1/D2); clean the shell and confirm no coating breach or active corrosion beneath the staining; recoat if required.
2. Close-range inspection of the lower mid-tower flange connection (Connection L1 D1) for corrosion, bolt condition and sealant; touch-up / recoat if paint breach or corrosion is confirmed.
3. Verify the base anchor-bolt ring (Base D2): bolt-cap condition, corrosion, torque marks and grout / base-seal integrity.
4. Verify circumferential weld and flange seams (all segments) for paint continuity and any coating cracking; log for trend.
5. Clean-and-reinspect the stained / scuffed shell panels to separate soiling and water-run staining from any coating breach.
6. Maintain a standardized periodic drone inspection for consistent trend tracking; re-image the tower top and flange connections next cycle to monitor corrosion-staining progression.